Simulation Analysis

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1 How to read this document

This document is divided into sections based on specific topics. In each section there is a set of summary main bullet points, a description of the analysis done in that section to support those bullet points, and the output from the analysis done in that section. The main bullet points cover a combination of answering the questions in the empirical paper and the overall questions about how each of the estimators perform under different conditions. The analysis descriptions need to be moved to be under the piece of analysis they describe, but for now I thought it was better to have it all up top.

2 Research Questions

- Which estimators produce the best estimates of the average treatment effect?
- Which estimators produce the best estimates of the treatment effect standard error?
- Are the standard errors targeting finite versus super population

Relevant section from the empirical paper:

We discuss this further in our simulations, but the consequence is that any of our standard error estimators can be viewed as conservative, if we view the clusters as a finite population, or as appropriate if we consider the clusters as coming from a larger superpopulation (Litschwartz et al., 2025).

- Do estimated standard errors vary across estimators and why?

Notes: Is this because true SEs vary across estimators? Is this estimation error? Are certain SE estimators more accurate than others?

Relevant section from the empirical paper: Choosing an estimand and excluding the fully interacted estimators results in estimated standards errors that are less variable across estimators. However, it is notable that differences are still often non-trivial. Even in Figure we see estimated standard errors are 20 percent different across estimators in 24 percent of the cases we examined.

- Are the fully-interacted models driving the instability of the estimated standard errors?

Relevant section from the empirical paper:

For example, the largest estimated standard error is 1.5x the smallest estimated standard error in only 10 percent of cases without the fully-interacted models, down from 44 percent when considering all estimators. The ratio never exceeds 1.8 once the fully-interacted models are excluded. This corroborates simulation findings that show the instability of estimated standard errors coming from fully-interacted estimators (Litschwartz et al., 202X). In general, we do not recommend using the fully-interacted estimators due to this short-coming.

- Are problems caused by low degrees of freedom due to estimate instability or bias?

Relevant section from the empirical paper: Notice that in Figure 2 it appears that choice of estimator matters more in low-degree of freedom studies – the ranges across estimators tend to be larger in low degree of freedom studies (blue dots) than high degree of freedom studies (red dots).

[ADD SOMETHING ABOUT WHAT WE LEARNED FROM THE SIMULATIONS – DO WE THINK THESE RANGES ARE MORE ABOUT BIAS OR ESTIMATION ERROR]

3 Data Generating Process

The simulations are a fully factored experiment with the following data generating parameters.

Table 1: Simulation Data Generating Parameters

Parameter	**Values**
Block-Level Fixed	
Number of Blocks	5
Average Treatment Effect	.2
Block-Level ICC	.2
Average Number of Observations Per Cluster	20
Number of Block Level Covariates	0
Block-Level Varying	
Block Type	Same, Messy, Reasonable
Average Number of Clusters per Block	5, 10, 20
Proportion of Clusters Treated	.5, .7
Cluster-Level Varying	
Correlation of Impact and Size of Cluster	0, .8
Cluster Level Treatment Effect Standard Deviation	0, .3
Cluster Level ICC	0, .2
How much does the size of clusters vary (coefficient of variation)?	0, .33, .6
Number of Cluster-Level Covariates	0, 1, 5 [R^2 Always .5]

Table 2: Block-Type Parameters

Parameter	Blocks All The Same	Blocks All Messy	Blocks All Reasonable
How much do number clusters per block vary (coefficient of variation)?	0 (none)	.85 (big)	.33 (average)
Block-Level Average Treatment Effect Standard Deviation	0	.6	.1
Correlation Between Treatment Effect and Cluster Size	0	.8	.2
Correlation Between Average Number of Observations Per Cluster and Total Number of Clusters	0	.8	.1
Correlation of Proportion of Clusters Treated per Block and Impact	0	.8	0
Correlation Between Outcome Intercept and Cluster Size	0	.8	.1

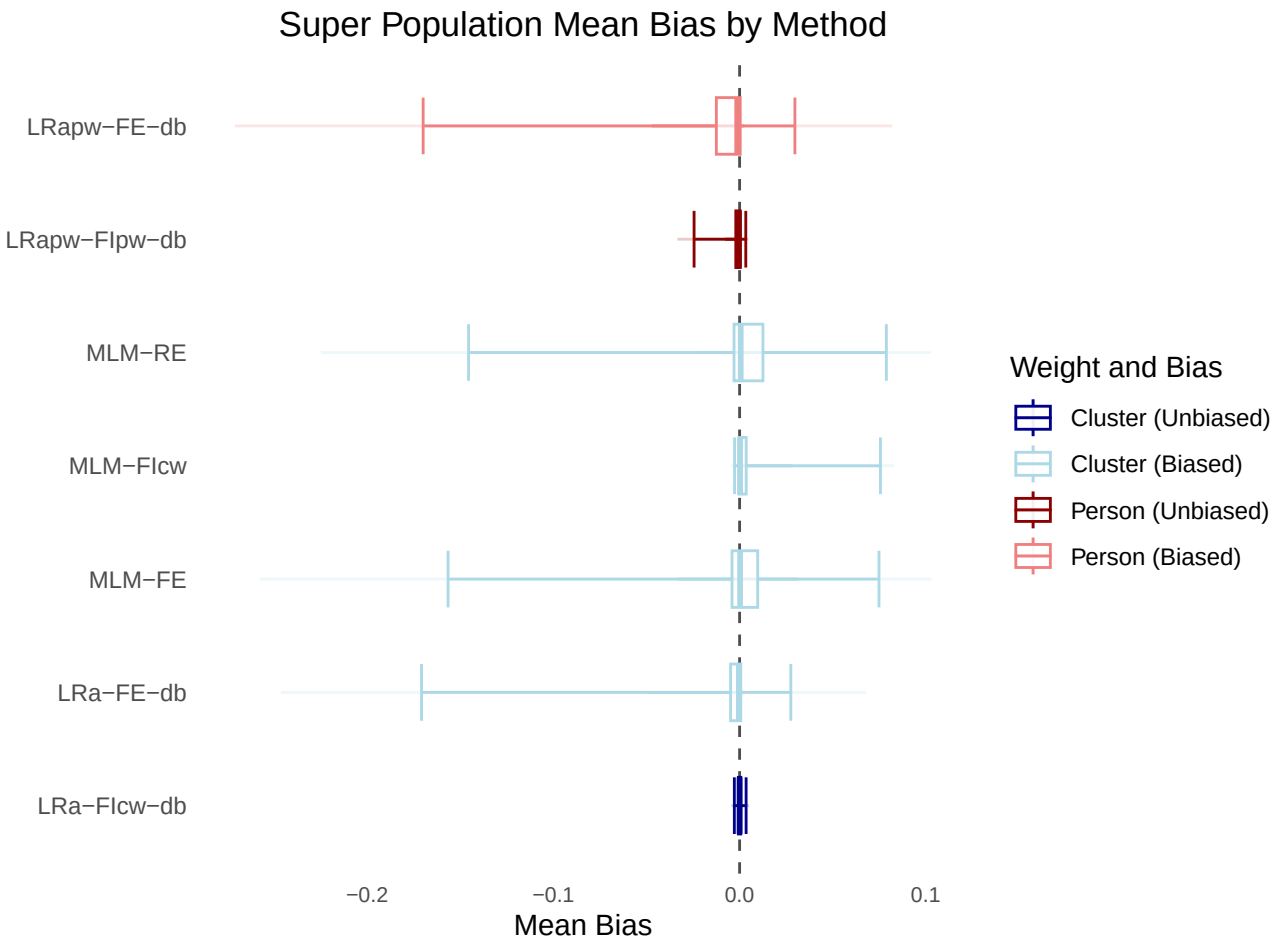
4 What does the bias look like for each estimator

4.1 Main Bullet Points:

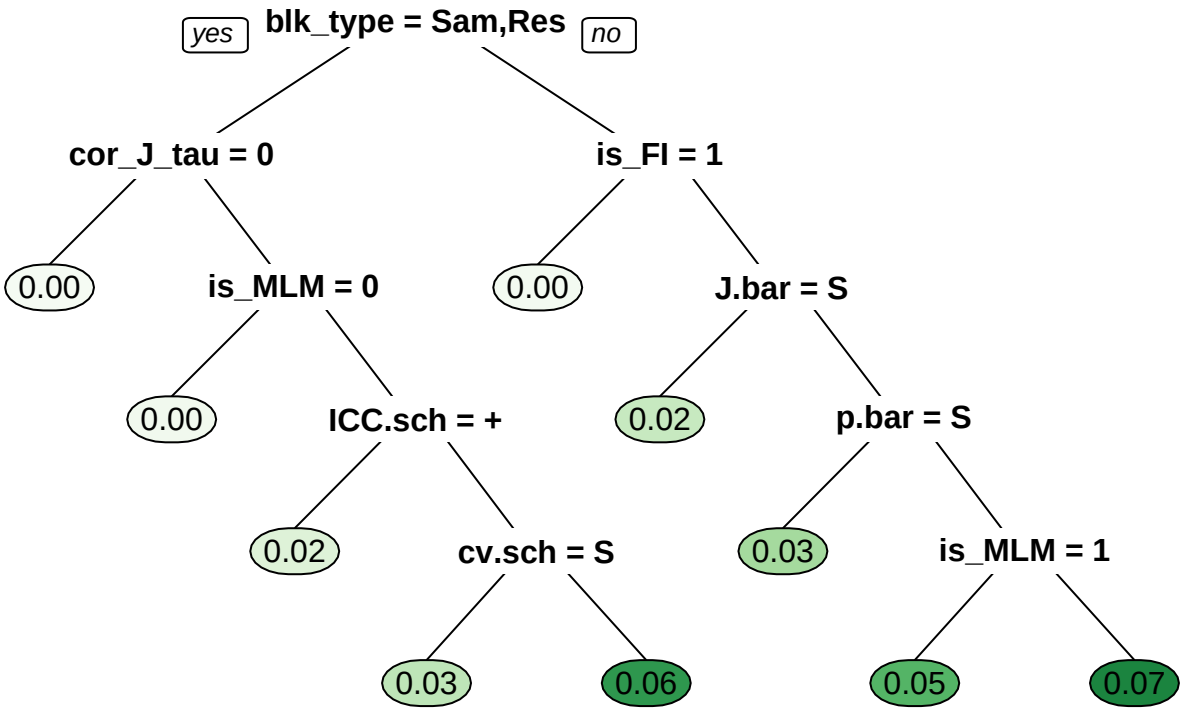
- The estimators theory says are unbiased do indeed have no bias (or have very low bias) across the full range of simulations
- For most of the estimators that are biased, the mean bias is mostly driven by large differences across blocks. With the largest bias in cases with large differences across blocks, 10 or 20 clusters, and unbalanced assignment to treatment (mean p =.7).
- The LRapw−FIpw−db has bias as large as -.033, which from the regression tree is an driven by a correlation between the treatment effect and cluster size and then variance in cluster size and treatment effect. This is consistent with theory.

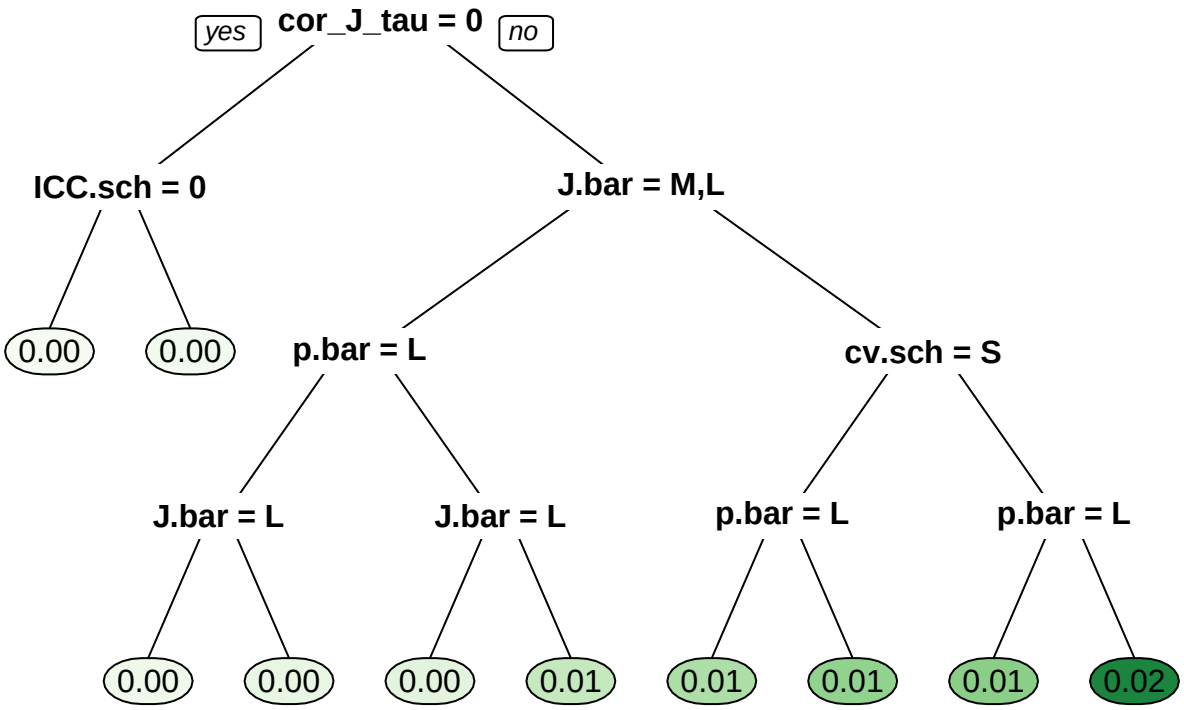
4.2 Description of Analysis

- The first graph is a boxplot of mean bias across the different scenarios by method. The faint horizontal lines go out to the full range, the bars are the 1% to 99%, and the boxes are the 10% to 90%.
- The second graph is a regression tree which predicts the magnitude of the mean bias using different data generating parameters and also flags for if the method is an MLM, is fully interacted, is cluster weighted, or is person weighted.
- The third graph is a the same regression tree above, but just for the LRapw−FIpw−db. The indicators for MLM, FI, cluster weighted, and person weighted are dropped for this regression.
- The first table contains the DGPs that produce that largest magnitude of mean bias for each method, conditional on the magnitude of the mean bias being at least .1 . Where again mean bias is the mean bias against the targeted estimator.


Super Population Mean Bias by Method
LRapw−FE−db
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRa−FE−db
LRa−FIcw−db
−0.2
−0.1
0.0
0.1
Mean Bias
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)

Decision Tree for Magnitude of Bias for Targeted Estimand





4.2.1 Top Five Scenarios with Most Bias by Method

method	bias.s estimand	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type	p.bar
LRa-FE-db	-0.2463	M	+	+	L	L	0	Mess	L
LRa-FE-db	-0.2278	M	+	0	L	0	+	Mess	L
LRa-FE-db	-0.2265	L	+	+	S	0	0	Mess	L
LRa-FE-db	-0.2246	M	0	0	0	S	0	Mess	L
LRa-FE-db	-0.2221	S	+	0	S	L	+	Mess	L
LRapw-FE-db	-0.2711	M	+	+	L	L	0	Mess	L
LRapw-FE-db	-0.2694	S	+	0	S	L	+	Mess	L
LRapw-FE-db	-0.2552	L	+	+	S	0	0	Mess	L

method	bias.s estimand	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type	p.bar
LRapw-FE-db	-0.2446	M	+	0	L	0	+	Mess	L
LRapw-FE-db	-0.2245	M	+	+	S	0	+	Mess	L
MLM-FE	-0.2576	M	+	+	L	L	0	Mess	L
MLM-FE	-0.2289	L	+	+	S	0	0	Mess	L
MLM-FE	-0.2029	S	+	0	S	L	+	Mess	L
MLM-FE	-0.1987	M	0	0	0	S	0	Mess	L
MLM-FE	-0.1984	M	0	0	0	L	0	Mess	L
MLM-FIcw	0.08312	L	+	0	L	L	+	Reas	S
MLM-FIcw	0.08233	L	+	0	L	S	+	Same	S
MLM-FIcw	0.08093	L	+	0	L	0	+	Same	S
MLM-FIcw	0.08051	L	+	0	L	L	+	Same	S
MLM-FIcw	0.07983	L	+	0	L	0	+	Reas	S
MLM-RE	-0.2249	M	+	+	L	L	0	Mess	L
MLM-RE	-0.2175	L	+	+	S	0	0	Mess	L
MLM-RE	-0.1922	M	0	0	0	L	0	Mess	L
MLM-RE	-0.1895	M	0	0	0	S	0	Mess	L
MLM-RE	-0.1796	S	+	0	S	L	+	Mess	L
LRapw-FIpw-db	-0.0334	S	+	+	L	0	+	Same	S
LRapw-FIpw-db	-0.03231	S	+	0	L	0	+	Same	S
LRapw-FIpw-db	-0.0321	S	+	0	L	L	+	Same	S
LRapw-FIpw-db	-0.03112	S	+	0	L	S	+	Same	S
LRapw-FIpw-db	-0.02909	S	+	+	L	S	+	Same	S

5 What does the standard error look like for each estimator

5.1 Main Bullet Points:

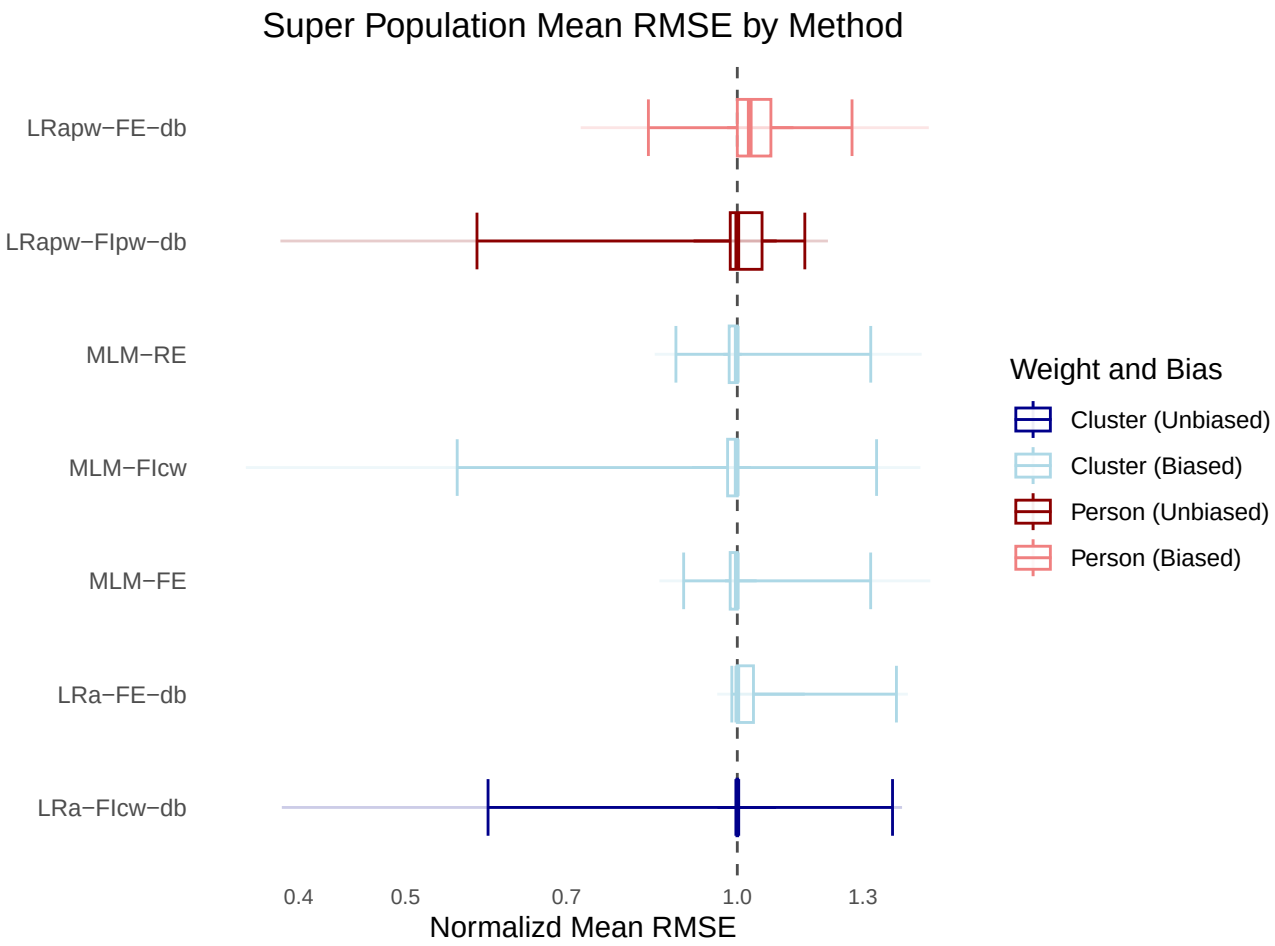
- For the person weighted estimators, the unbiased estimator both has less error and is more stable than the biased estimator.
- For the cluster weighted estimators, the unbiased estimator generally has less error than the biased estimators but more instablity.
- The MLM estimators are the most stable, which fits with theory.
- High RMSE and estimate instability is driven by a cluster ICC and a low number of clusters

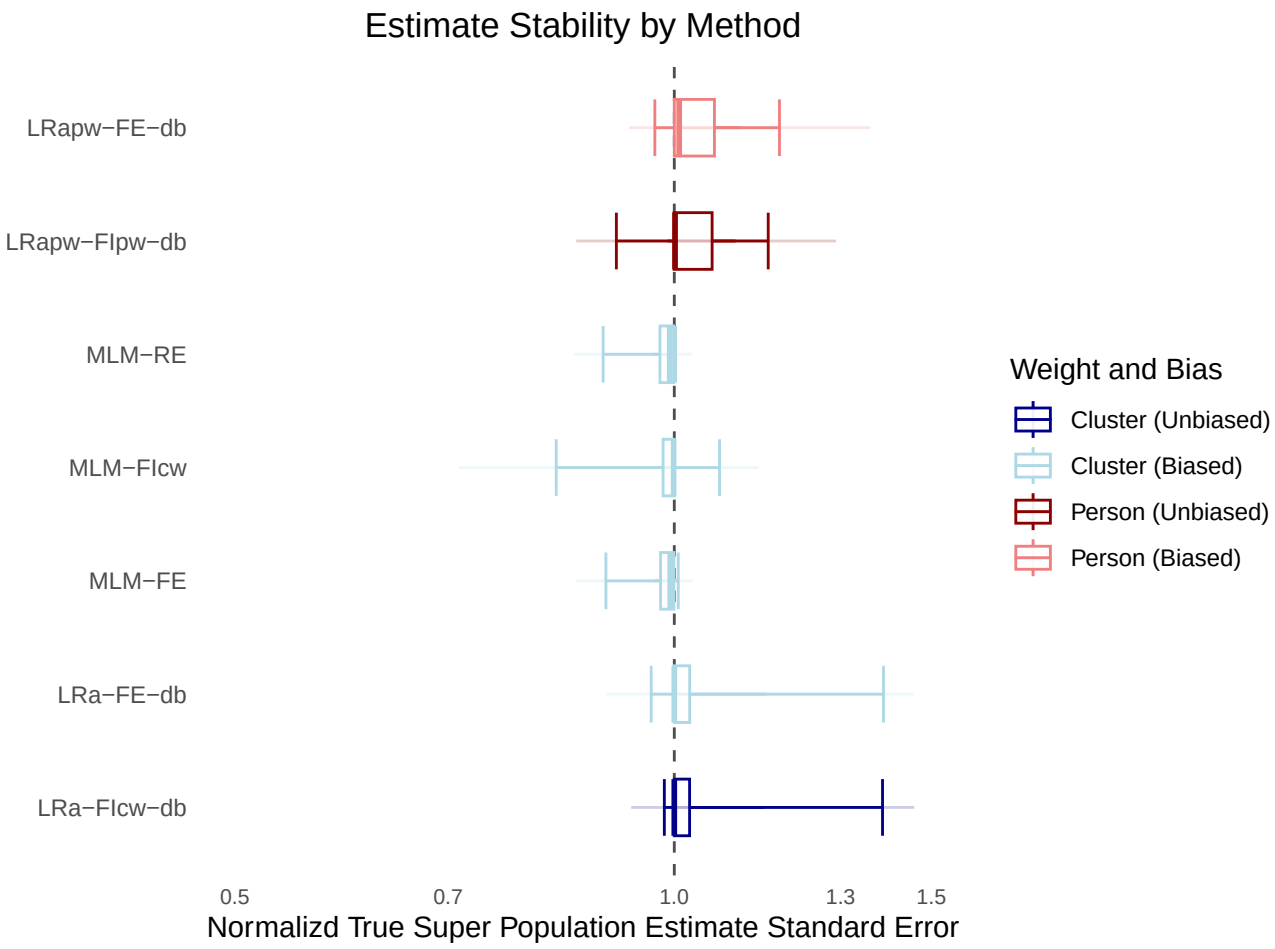
5.2 Answers to Questions In the Emperical Paper

- Re the question on pg. 39, looking at the second boxplot shows that true S.E. do vary across estimators with the true standard error in a given DGP often 40% bigger for some estimates than the median estimate in that DPG.

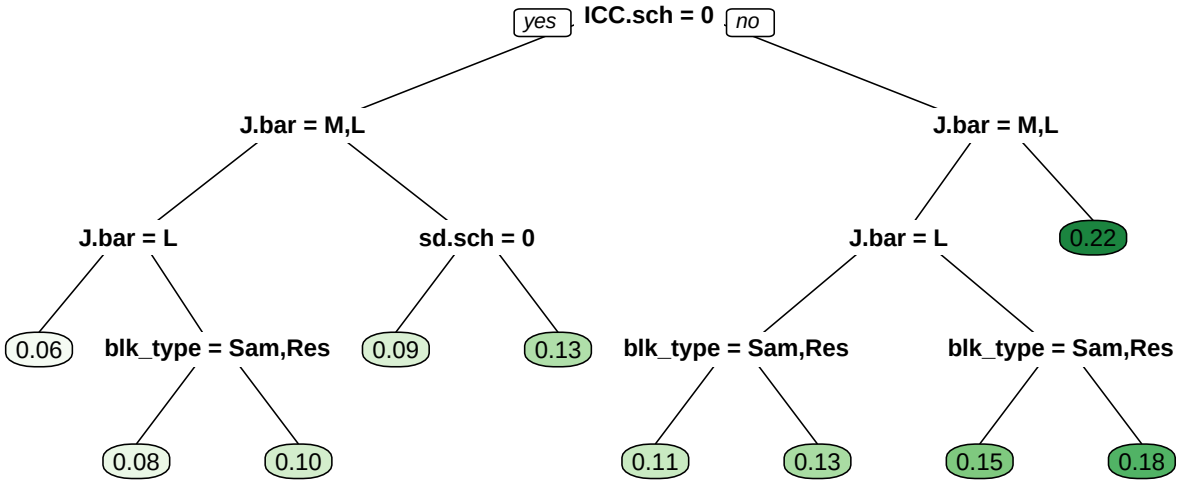
5.3 Description of Analysis:

- The first graph is a boxplot of the normalized RMSE across the different scenarios by method. The normalized RMSE is the RMSE for given method divided by the median RMSE across methods within a given simulation scenario. Therefore each data point is adjusted for the underlying base RMSE from a given set of DGPs and represents the RMSE driven by the estimation method.
- The second graph is a boxplot of the normalized estimate standard deviation across the different scenarios by method. This is also normalized by dividing by the median estimate standard deviation within a given simulation scenario.
- The third graph is a regression tree which predicts the root mean squared error (RMSE) against the targeted estimand using different data generating parameters and also flags for if the method is an MLM, is fully interacted, is cluster weighted, or is person weighted.
- The fourth graph is a regression tree which predicts the true super population standard error using different data generating parameters and also flags for if the method is an MLM, is fully interacted, is cluster weighted, or is person weighted.
- The first table contains the five DGPs that produce the five most normalized RMSE (against the targeted estimator) for each method, conditional on the magnitude of the being at least 1.10.
- The fourth table contains the five DGPs that produce the most normalized super population standard error for each method, conditional on the magnitude of the being at least 1.10.

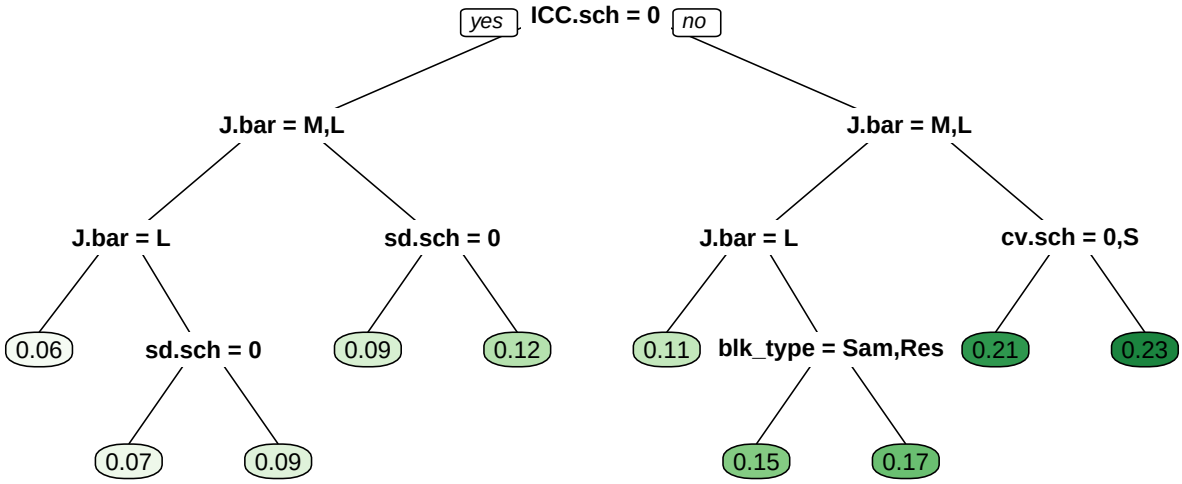

Super Population Mean RMSE by Method
LRapw−FE−db
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRa−FE−db
LRa−FIcw−db
0.4
0.5
0.7
1.0
1.3
Normalizd Mean RMSE
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)


Estimate Stability by Method
LRapw−FE−db
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRa−FE−db
LRa−FIcw−db
0.5
0.7
1.0
1.3
1.5
Normalizd True Super Population Estimate Standard Error
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)

Decision Tree for RMSE Against Targeted Estimand



Decision Tree for Super Population True Standard Error



5.3.1 Top Five Scenarios with Largest Normalized RMSE by Method

method	RMSE.s.estimand.norm	RMSE.s estimand	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	1.79	0.104	L	0	0	0	S	0	Mess
LRa-FE-db	1.57	0.253	M	+	0	L	0	+	Mess
LRa-FE-db	1.52	0.104	L	0	0	0	L	0	Mess
LRa-FE-db	1.43	0.161	M	0	0	L	0	0	Mess
LRa-FE-db	1.42	0.101	L	0	0	S	L	0	Mess
LRapw-FE-db	1.87	0.109	L	0	0	0	S	0	Mess
LRapw-FE-db	1.72	0.278	M	+	0	L	0	+	Mess
LRapw-FE-db	1.49	0.179	L	+	0	L	L	+	Mess

method	RMSE.s.estimand.norm	RMSE.sestimand	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRapw-FE-db	1.41	0.296	S	+	0	S	L	+	Mess
LRapw-FE-db	1.4	0.149	M	+	0	L	S	+	Mess
MLM-FE	1.86	0.108	L	+	0	L	0	0	Mess
MLM-FE	1.63	0.12	L	+	0	L	0	+	Reas
MLM-FE	1.53	0.0534	L	0	0	0	0	0	Mess
MLM-FE	1.5	0.0806	L	0	0	S	L	0	Mess
MLM-FE	1.47	0.106	L	+	0	L	0	+	Reas
MLM-FIcw	1.47	0.101	L	+	0	L	L	+	Reas
MLM-FIcw	1.46	0.0983	L	+	0	L	L	+	Same
MLM-FIcw	1.45	0.0999	L	+	0	L	S	+	Same
MLM-FIcw	1.45	0.0988	L	+	0	L	0	+	Same
MLM-FIcw	1.37	0.101	L	+	0	L	0	+	Reas
MLM-RE	1.84	0.107	L	+	0	L	0	0	Mess
MLM-RE	1.63	0.12	L	+	0	L	0	+	Reas
MLM-RE	1.52	0.0817	L	0	0	S	L	0	Mess
MLM-RE	1.51	0.0527	L	0	0	0	0	0	Mess
MLM-RE	1.47	0.106	L	+	0	L	0	+	Reas
LRa-FIcw-db	1.41	0.0568	L	0	0	L	0	0	Same
LRa-FIcw-db	1.41	0.0573	L	0	0	L	L	0	Same
LRa-FIcw-db	1.4	0.0565	L	0	0	L	S	0	Same
LRa-FIcw-db	1.4	0.0618	L	0	0	L	L	0	Same
LRa-FIcw-db	1.4	0.062	L	0	0	L	0	0	Same
LRapw-FIpw-db	1.21	0.295	S	+	+	L	S	+	Mess
LRapw-FIpw-db	1.18	0.126	L	+	+	L	S	0	Reas

method	RMSE.s.est imand.norm	RMSE.s estimand	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRapw-FIpw-db	1.17	0.24	S	+	+	L	L	+	Reas
LRapw-FIpw-db	1.16	0.255	S	+	+	L	L	0	Reas
LRapw-FIpw-db	1.16	0.285	S	+	+	L	L	+	Mess

5.3.2 Top Five Scenarios with Largest Normalized Estimate Standard Error by Method

method	SE s.norm	SE.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	1.46	0.107	S	0	0	L	L	0	Reas
LRa-FE-db	1.43	0.0541	L	0	0	L	S	0	Reas
LRa-FE-db	1.41	0.0567	L	0	0	L	0	0	Same
LRa-FE-db	1.41	0.0573	L	0	0	L	L	0	Same
LRa-FE-db	1.4	0.0565	L	0	0	L	S	0	Same
LRapw-FE-db	1.36	0.0784	L	0	0	L	S	0	Mess
LRapw-FE-db	1.24	0.132	M	+	0	L	0	+	Mess
LRapw-FE-db	1.23	0.302	S	+	+	L	S	+	Mess
LRapw-FE-db	1.21	0.133	S	+	0	L	S	0	Mess
LRapw-FE-db	1.21	0.277	S	+	+	L	L	0	Mess
MLM-FIcw	1.14	0.0984	S	0	0	0	L	0	Reas
MLM-FIcw	1.14	0.0685	M	0	0	S	S	0	Reas
MLM-FIcw	1.13	0.0576	L	0	0	S	L	0	Reas
MLM-FIcw	1.12	0.0822	S	0	0	L	L	0	Reas
MLM-FIcw	1.11	0.06	M	0	0	0	0	0	Reas
LRa-FIcw-db	1.46	0.107	S	0	0	L	L	0	Reas
LRa-FIcw-db	1.43	0.0541	L	0	0	L	S	0	Reas

method	SE s.norm	SE.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FIcw-db	1.41	0.0567	L	0	0	L	0	0	Same
LRa-FIcw-db	1.41	0.0573	L	0	0	L	L	0	Same
LRa-FIcw-db	1.4	0.0565	L	0	0	L	S	0	Same
LRapw-FIpw-db	1.29	0.256	M	+	+	S	S	+	Mess
LRapw-FIpw-db	1.21	0.13	L	0	+	L	0	0	Mess
LRapw-FIpw-db	1.2	0.294	S	+	+	L	S	+	Mess
LRapw-FIpw-db	1.18	0.126	L	+	+	L	S	0	Reas
LRapw-FIpw-db	1.17	0.116	L	0	+	L	L	0	Mess

6 What do the estimated standard errors look like for each estimator

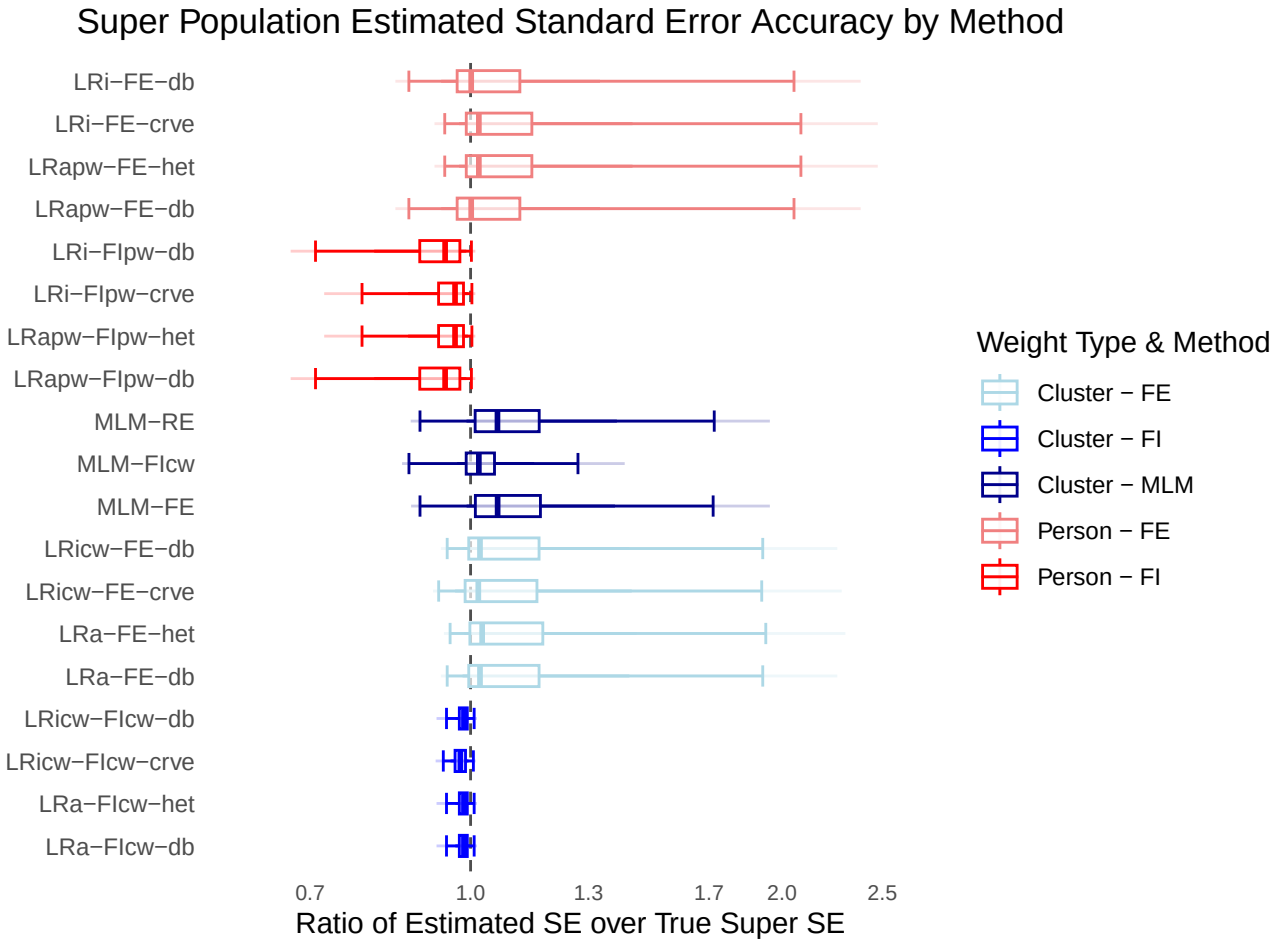
6.1 Main Bullet Points:

- This section answers the rest of the questions from pg.39 (i.e. are certain SE estimators more accurate than others)
- For the cluster targeted estimates the fully interacted estimators produce more accurate standard errors than the other estimators (graph one). The unbiased estimators produce more unstable estimates than the biased estimators, but in 90% of the DGP we looked at they aren't a lot more unstable (graph four).
- For the person targeted estimates the fully interacted estimators produce more accurate standard errors than the other estimators (graphs one, two and three). The fully interacted estimators produce more unstable estimates than the other estimators (graph four). However they aren't a lot more unstable, 90% of the time they're within about 10% of the median instability and fully interacted doesn't come up in the regression tree (graph five).
- This section also answers the question from pg. 38 about the instability of the estimated standard errors for the fully interacted models. The simulations do not really show that the standard errors from the fully interacted models are lot more unstable than the other standard errors. We should talk about this more, and what would be helpful for me to dig into more.

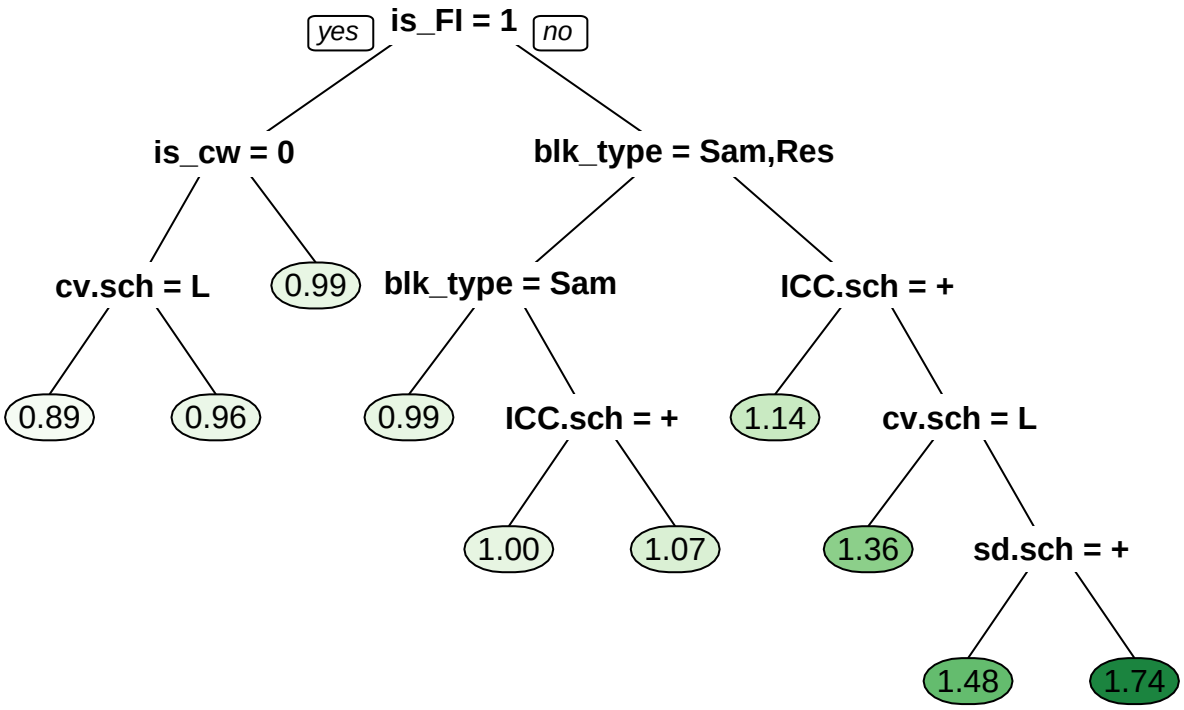
6.2 Description of Analysis:

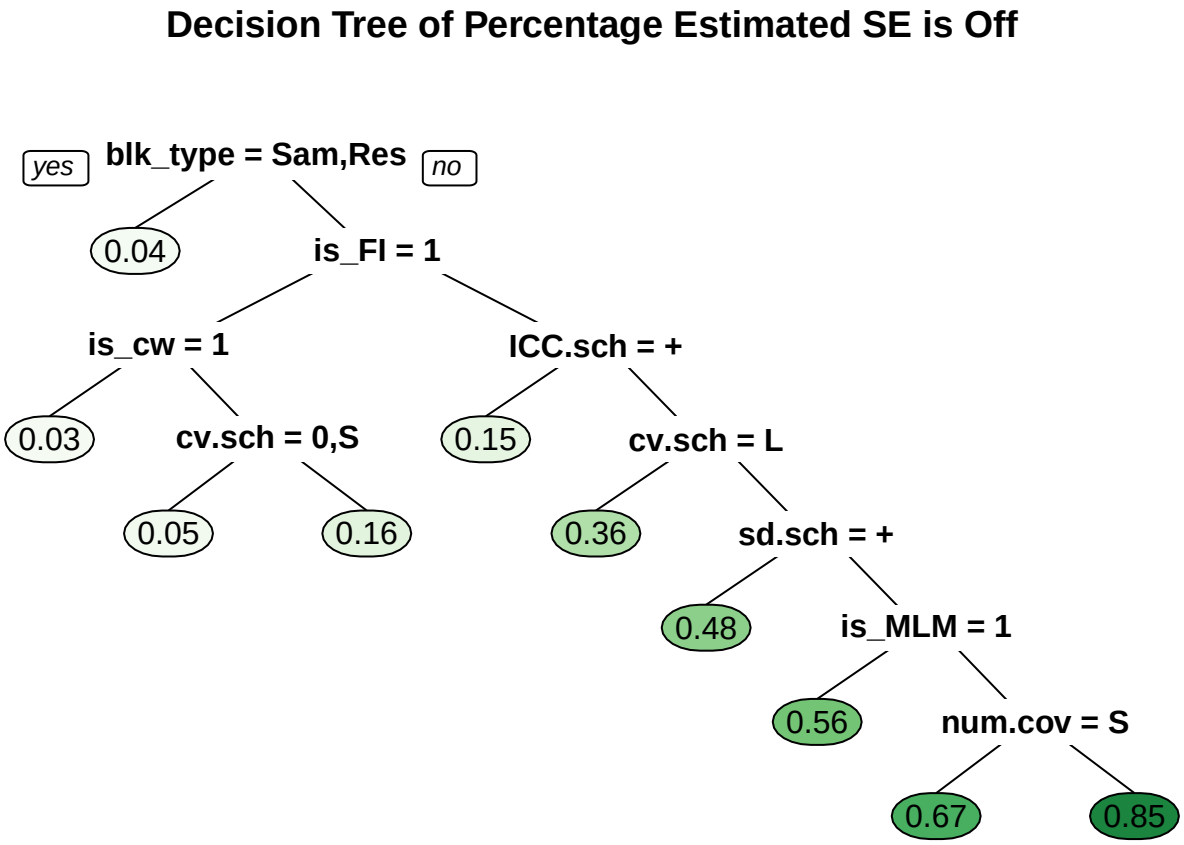
- The first graph is a boxplot of the estimated standard error accuracy across the different scenarios by method. Standard error accuracy is quantified as the estimated standard error over the true super population standard error. Estimated standard errors are most accurate if the number is close to one. Numbers below one represent under estimated standard errors and numbers above one represent over estimated standard errors. [Luke had a comment on this graph that I couldn't read and need to follow up on]

- The second graph is a regression tree which predicts the standard error inflation (estimated standard error over the true super population standard error) using different data generating parameters and also flags for if the method is an MLM, is fully interacted, is cluster weighted, or is person weighted.
- For the third graph, I was having trouble conceptualizing the scale of standard error inflation, so I made a regression tree where I converted it to percentage inaccuracy. The regression tree here is therefore predicting the magnitude of ((Mean Estimated SE - True Super Population SE) / True Super Population SE) * 100.
- The first table contains the five DGPs that produce the most inaccurate standard error estimates for each method, conditional on the magnitude of the inaccuracy being at least .1.
- The fourth graph is a boxplot of the standard deviation of the estimated standard error across the different replications divided by the true estimate standard error by method.
- The fifth graph is a boxplot of the normalized standard deviation of the estimated standard error across the different replications by method.
- The sixth graph is a regression tree which predicts instability in the standard error estimate using different data generating parameters and also flags for if the method is an MLM, is fully interacted, is cluster weighted, or is person weighted. Here instability in the standard error estimate is quantified as the standard deviation of the estimated standard error across the different replications divided by the true estimate standard error by method
- The second table contains the five DGPs that produce the largest standard deviation in the standard error estimates for each method, conditional on the magnitude being at least .05.
- The third table contains the five DGPs that produce the largest normalized standard deviation in the standard error estimates for each method, conditional on the magnitude being at least 1.10.


Super Population Estimated Standard Error Accuracy by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
0.7
1.0
1.3
1.7
2.0
2.5
Ratio of Estimated SE over True Super SE
Weight Type & Method
Cluster − FE
Cluster − FI
Cluster − MLM
Person − FE
Person − FI

Decision Tree of Ratio of Estimated SE over True Super SE



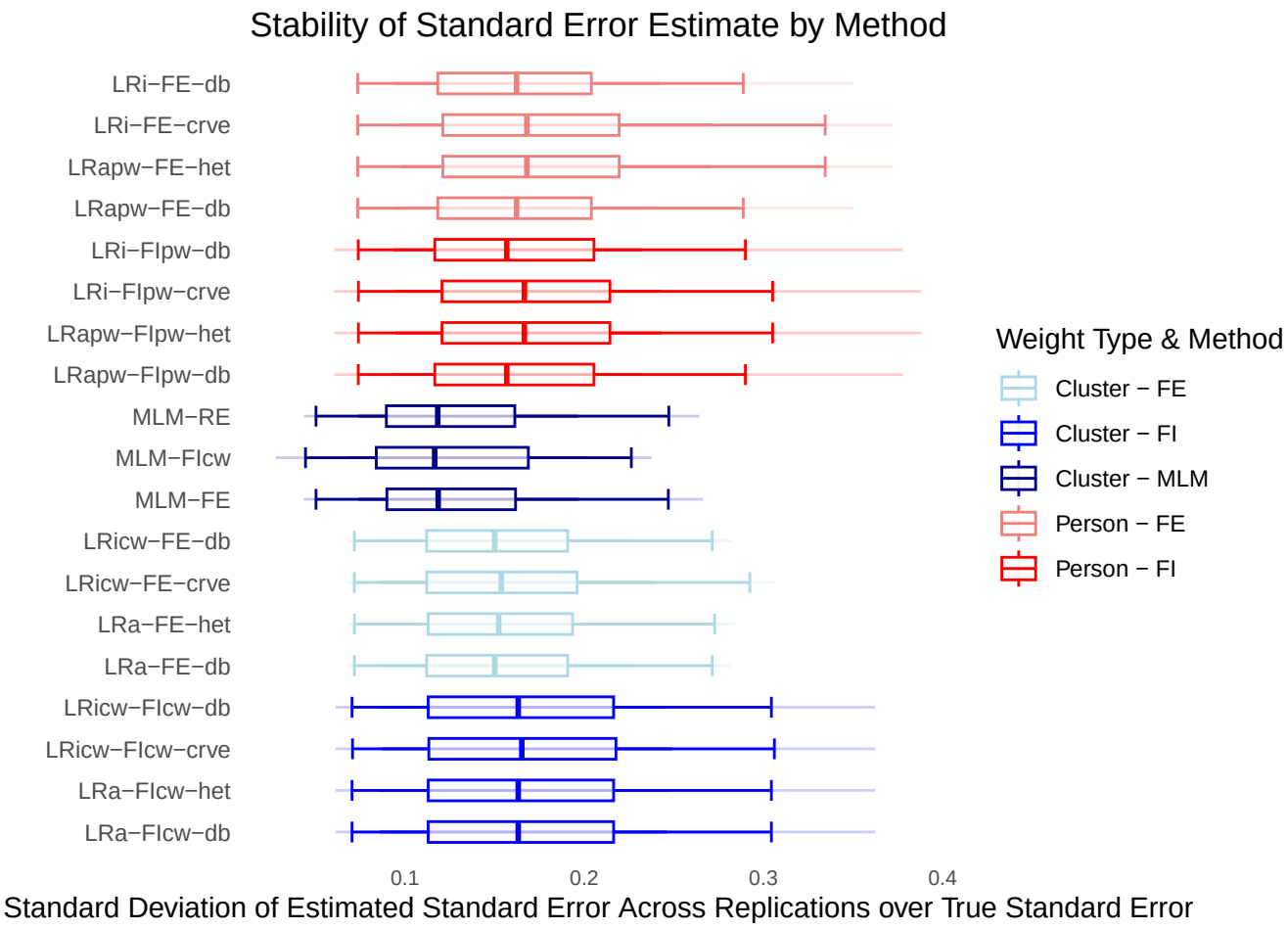


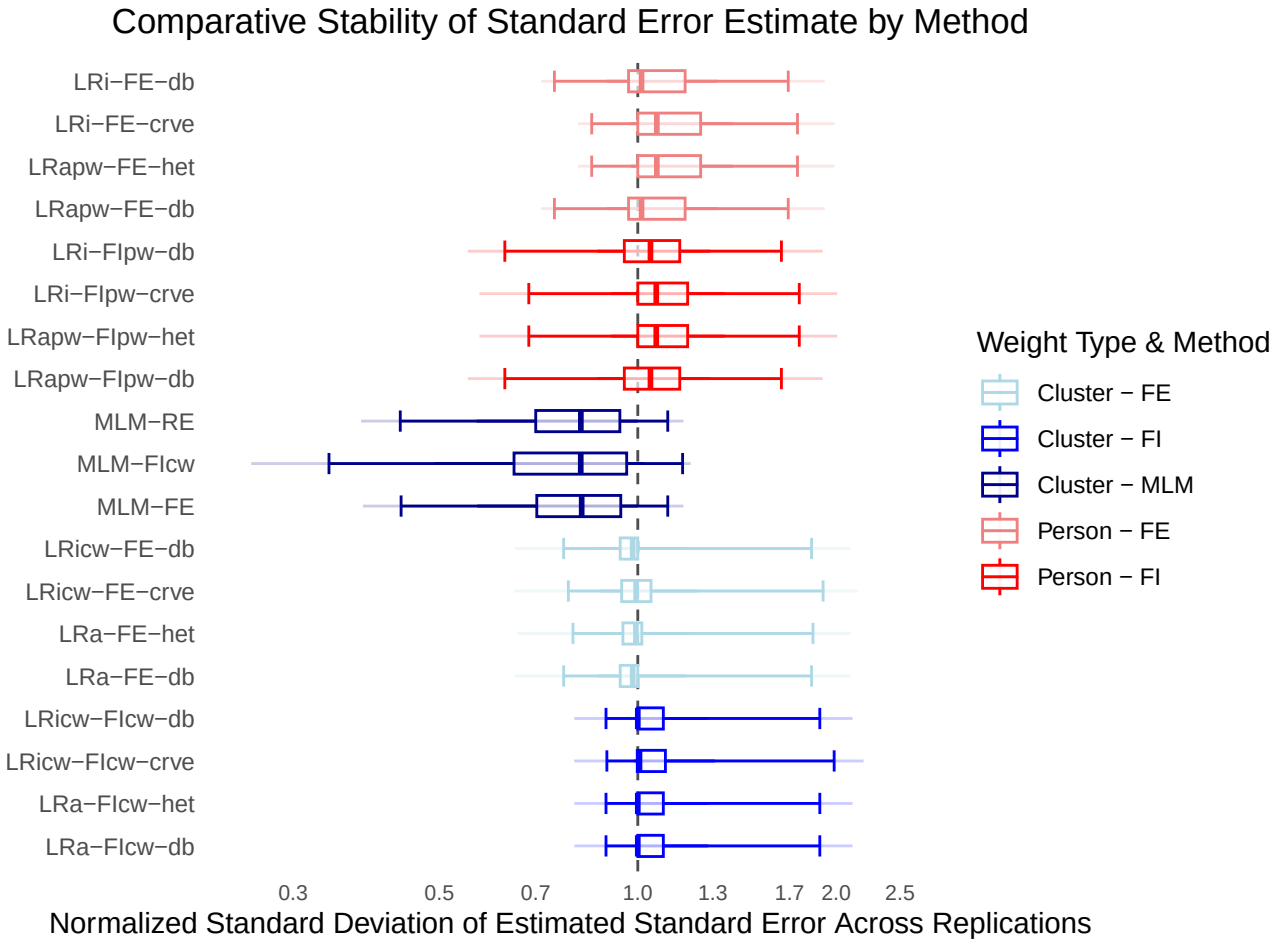
6.2.1 Top Five Scenarios with Most SE Estimate Inaccuracy by Method

method	infla te.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	2.26	L	0	0	0	0	0	Mess
LRa-FE-db	2.21	M	0	0	S	0	0	Mess
LRa-FE-db	2.21	S	0	0	0	L	0	Mess
LRa-FE-db	2.15	M	0	0	S	L	0	Mess
LRa-FE-db	2.15	M	0	0	0	S	0	Mess
LRa-FE-het	2.3	S	0	0	0	L	0	Mess
LRa-FE-het	2.29	L	0	0	0	0	0	Mess
LRa-FE-het	2.23	M	0	0	S	0	0	Mess

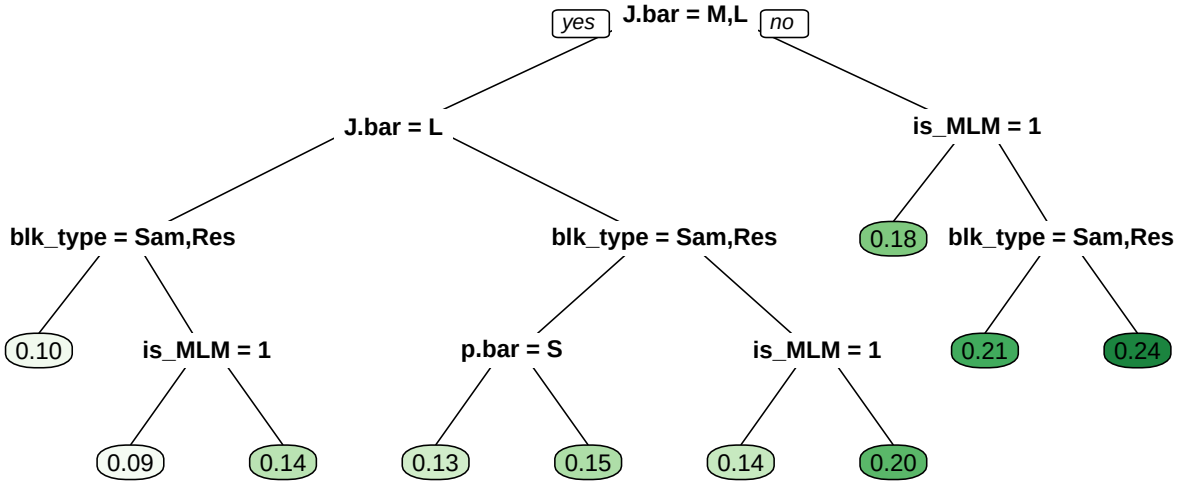
method	infla te.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-het	2.21	M	0	0	S	L	0	Mess
LRa-FE-het	2.2	M	0	0	0	S	0	Mess
LRapw-FE-db	2.38	L	0	0	0	0	0	Mess
LRapw-FE-db	2.37	S	0	0	0	L	0	Mess
LRapw-FE-db	2.17	M	0	0	0	L	0	Mess
LRapw-FE-db	2.13	M	0	0	S	L	0	Mess
LRapw-FE-db	2.13	M	0	0	S	0	0	Mess
LRapw-FE-het	2.47	S	0	0	0	L	0	Mess
LRapw-FE-het	2.41	L	0	0	0	0	0	Mess
LRapw-FE-het	2.26	M	0	0	S	L	0	Mess
LRapw-FE-het	2.2	M	0	0	0	L	0	Mess
LRapw-FE-het	2.19	M	0	0	S	0	0	Mess
LRicw-FE-crve	2.28	L	0	0	0	0	0	Mess
LRicw-FE-crve	2.28	S	0	0	0	L	0	Mess
LRicw-FE-crve	2.21	M	0	0	S	0	0	Mess
LRicw-FE-crve	2.19	M	0	0	0	S	0	Mess
LRicw-FE-crve	2.18	M	0	0	0	L	0	Mess
MLM-FE	1.95	S	0	0	0	0	0	Mess
MLM-FE	1.87	S	0	0	0	0	0	Mess
MLM-FE	1.85	S	0	0	0	L	0	Mess
MLM-FE	1.79	S	0	0	S	0	0	Mess
MLM-FE	1.78	S	0	0	S	L	0	Mess
MLM-FIcw	1.41	S	+	0	S	L	+	Mess
MLM-FIcw	1.38	M	+	0	S	0	+	Mess
MLM-FIcw	1.38	L	+	0	0	S	0	Mess
MLM-FIcw	1.38	L	+	0	S	0	+	Mess
MLM-FIcw	1.34	L	+	0	L	S	0	Mess
MLM-RE	1.95	S	0	0	0	0	0	Mess
MLM-RE	1.87	S	0	0	0	0	0	Mess

method	infla te.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
MLM-RE	1.85	S	0	0	0	L	0	Mess
MLM-RE	1.8	S	0	0	S	0	0	Mess
MLM-RE	1.78	S	0	0	S	0	0	Mess
LRapw-FIpw-db	0.67	S	+	0	L	0	+	Mess
LRapw-FIpw-db	0.682	S	0	0	L	S	0	Mess
LRapw-FIpw-db	0.694	L	0	0	L	S	0	Mess
LRapw-FIpw-db	0.695	S	+	0	L	0	0	Mess
LRapw-FIpw-db	0.704	S	0	0	L	0	0	Mess
LRapw-FIpw-het	0.722	L	0	0	L	S	0	Mess
LRapw-FIpw-het	0.749	S	+	0	L	0	+	Mess
LRapw-FIpw-het	0.767	S	+	+	L	S	+	Mess
LRapw-FIpw-het	0.767	S	+	0	L	0	0	Mess
LRapw-FIpw-het	0.769	S	0	0	L	S	0	Mess


Stability of Standard Error Estimate by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
0.1
0.2
0.3
0.4
Standard Deviation of Estimated Standard Error Across Replications over True Standard Error
Weight Type & Method
Cluster − FE
Cluster − FI
Cluster − MLM
Person − FE
Person − FI


Comparative Stability of Standard Error Estimate by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
0.3
0.5
0.7
1.0
1.3
1.7
2.0
2.5
Normalized Standard Deviation of Estimated Standard Error Across Replications
Weight Type & Method
Cluster − FE
Cluster − FI
Cluster − MLM
Person − FE
Person − FI

Decision Tree of Ratio of SD of Estimated SE over True Super SE



6.2.2 Top Five Scenarios with Most SE Estimate Instability by Method

method	infla te.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	2.26	L	0	0	0	0	0	Mess
LRa-FE-db	2.21	M	0	0	S	0	0	Mess
LRa-FE-db	2.21	S	0	0	0	L	0	Mess
LRa-FE-db	2.15	M	0	0	S	L	0	Mess
LRa-FE-db	2.15	M	0	0	0	S	0	Mess
LRa-FE-het	2.3	S	0	0	0	L	0	Mess
LRa-FE-het	2.29	L	0	0	0	0	0	Mess
LRa-FE-het	2.23	M	0	0	S	0	0	Mess

method	infla te.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-het	2.21	M	0	0	S	L	0	Mess
LRa-FE-het	2.2	M	0	0	0	S	0	Mess
LRapw-FE-db	2.38	L	0	0	0	0	0	Mess
LRapw-FE-db	2.37	S	0	0	0	L	0	Mess
LRapw-FE-db	2.17	M	0	0	0	L	0	Mess
LRapw-FE-db	2.13	M	0	0	S	L	0	Mess
LRapw-FE-db	2.13	M	0	0	S	0	0	Mess
LRapw-FE-het	2.47	S	0	0	0	L	0	Mess
LRapw-FE-het	2.41	L	0	0	0	0	0	Mess
LRapw-FE-het	2.26	M	0	0	S	L	0	Mess
LRapw-FE-het	2.2	M	0	0	0	L	0	Mess
LRapw-FE-het	2.19	M	0	0	S	0	0	Mess
LRicw-FE-crve	2.28	L	0	0	0	0	0	Mess
LRicw-FE-crve	2.28	S	0	0	0	L	0	Mess
LRicw-FE-crve	2.21	M	0	0	S	0	0	Mess
LRicw-FE-crve	2.19	M	0	0	0	S	0	Mess
LRicw-FE-crve	2.18	M	0	0	0	L	0	Mess
MLM-FE	1.95	S	0	0	0	0	0	Mess
MLM-FE	1.87	S	0	0	0	0	0	Mess
MLM-FE	1.85	S	0	0	0	L	0	Mess
MLM-FE	1.79	S	0	0	S	0	0	Mess
MLM-FE	1.78	S	0	0	S	L	0	Mess
MLM-FIcw	1.41	S	+	0	S	L	+	Mess
MLM-FIcw	1.38	M	+	0	S	0	+	Mess
MLM-FIcw	1.38	L	+	0	0	S	0	Mess
MLM-FIcw	1.38	L	+	0	S	0	+	Mess
MLM-FIcw	1.34	L	+	0	L	S	0	Mess
MLM-RE	1.95	S	0	0	0	0	0	Mess
MLM-RE	1.87	S	0	0	0	0	0	Mess

method	infla te.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
MLM-RE	1.85	S	0	0	0	L	0	Mess
MLM-RE	1.8	S	0	0	S	0	0	Mess
MLM-RE	1.78	S	0	0	S	0	0	Mess
LRapw-FIpw-db	0.67	S	+	0	L	0	+	Mess
LRapw-FIpw-db	0.682	S	0	0	L	S	0	Mess
LRapw-FIpw-db	0.694	L	0	0	L	S	0	Mess
LRapw-FIpw-db	0.695	S	+	0	L	0	0	Mess
LRapw-FIpw-db	0.704	S	0	0	L	0	0	Mess
LRapw-FIpw-het	0.722	L	0	0	L	S	0	Mess
LRapw-FIpw-het	0.749	S	+	0	L	0	+	Mess
LRapw-FIpw-het	0.767	S	+	+	L	S	+	Mess
LRapw-FIpw-het	0.767	S	+	0	L	0	0	Mess
LRapw-FIpw-het	0.769	S	0	0	L	S	0	Mess

6.2.3 Top Five Scenarios with Most Normalized SE Estimate Instability by Method

method	sd.SE hat.s.norm	sd.SE hat.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	2.1	0.00768	L	0	0	L	S	0	Same
LRa-FE-db	2.07	0.00782	L	0	0	L	L	0	Same
LRa-FE-db	2.05	0.00785	L	0	0	L	0	0	Same
LRa-FE-db	2.02	0.00739	L	0	0	L	S	0	Reas
LRa-FE-db	1.87	0.00831	L	0	0	L	L	0	Reas
LRa-FE-het	2.1	0.00768	L	0	0	L	S	0	Same
LRa-FE-het	2.07	0.00782	L	0	0	L	L	0	Same
LRa-FE-het	2.05	0.00785	L	0	0	L	0	0	Same
LRa-FE-het	2.03	0.0074	L	0	0	L	S	0	Reas
LRa-FE-het	1.88	0.00832	L	0	0	L	L	0	Reas

method	sd.SE hat.s.norm	sd.SE hat.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRapw-FE-db	1.92	0.0202	L	+	0	S	0	0	Mess
LRapw-FE-db	1.8	0.018	L	0	+	L	L	0	Mess
LRapw-FE-db	1.79	0.0188	L	0	+	S	S	0	Reas
LRapw-FE-db	1.78	0.0231	L	+	+	L	0	+	Reas
LRapw-FE-db	1.78	0.0236	L	0	+	L	0	0	Mess
LRapw-FE-het	1.99	0.0208	L	+	0	S	0	0	Mess
LRapw-FE-het	1.86	0.105	S	+	+	L	S	+	Mess
LRapw-FE-het	1.85	0.0259	M	+	+	L	S	0	Reas
LRapw-FE-het	1.84	0.0239	L	+	+	L	0	+	Reas
LRapw-FE-het	1.81	0.0181	L	0	+	L	L	0	Mess
LRicw-FE-crve	2.16	0.00788	L	0	0	L	S	0	Same
LRicw-FE-crve	2.12	0.00803	L	0	0	L	L	0	Same
LRicw-FE-crve	2.1	0.00806	L	0	0	L	0	0	Same
LRicw-FE-crve	2.07	0.00758	L	0	0	L	S	0	Reas
LRicw-FE-crve	1.94	0.0161	M	0	0	L	0	0	Same
MLM-FE	1.17	0.00503	L	+	0	0	L	0	Same
MLM-FE	1.17	0.0051	L	+	0	0	S	0	Same
MLM-FE	1.17	0.00508	L	+	0	0	0	0	Same
MLM-FE	1.13	0.00536	L	+	0	S	0	0	Same
MLM-FE	1.11	0.00531	L	+	0	S	L	+	Same
MLM-FIcw	1.2	0.00522	L	+	0	0	0	0	Same
MLM-FIcw	1.2	0.00523	L	+	0	0	S	0	Same
MLM-FIcw	1.2	0.00515	L	+	0	0	L	0	Same
MLM-FIcw	1.18	0.0111	M	+	0	0	0	0	Same
MLM-FIcw	1.18	0.0232	S	+	0	S	S	+	Same
MLM-RE	1.17	0.00503	L	+	0	0	L	0	Same
MLM-RE	1.17	0.0051	L	+	0	0	S	0	Same
MLM-RE	1.17	0.00508	L	+	0	0	0	0	Same
MLM-RE	1.13	0.00536	L	+	0	S	0	0	Same
MLM-RE	1.11	0.00531	L	+	0	S	L	+	Same
LRa-FIcw-db	2.12	0.00774	L	0	0	L	S	0	Same

method	sd.SE hat.s.norm	sd.SE hat.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FIcw-db	2.08	0.00761	L	0	0	L	S	0	Reas
LRa-FIcw-db	2.08	0.00786	L	0	0	L	L	0	Same
LRa-FIcw-db	2.06	0.00791	L	0	0	L	0	0	Same
LRa-FIcw-db	1.93	0.00856	L	0	0	L	L	0	Reas
LRapw-FIpw-db	1.91	0.0253	L	0	+	L	0	0	Mess
LRapw-FIpw-db	1.87	0.0859	S	+	+	0	0	0	Mess
LRapw-FIpw-db	1.85	0.0968	M	+	+	S	S	+	Mess
LRapw-FIpw-db	1.81	0.019	L	0	+	S	S	0	Reas
LRapw-FIpw-db	1.76	0.0246	M	+	+	L	S	0	Reas
LRapw-FIpw-het	2.01	0.0266	L	0	+	L	0	0	Mess
LRapw-FIpw-het	1.92	0.0268	M	+	+	L	S	0	Reas
LRapw-FIpw-het	1.9	0.0994	M	+	+	S	S	+	Mess
LRapw-FIpw-het	1.88	0.0197	L	0	+	S	S	0	Reas
LRapw-FIpw-het	1.87	0.0188	L	+	+	L	S	0	Reas
LRicw-FIcw-crve	2.2	0.00804	L	0	0	L	S	0	Same
LRicw-FIcw-crve	2.16	0.00789	L	0	0	L	S	0	Reas
LRicw-FIcw-crve	2.16	0.00817	L	0	0	L	L	0	Same
LRicw-FIcw-crve	2.14	0.00822	L	0	0	L	0	0	Same
LRicw-FIcw-crve	2.02	0.0115	L	0	0	L	S	0	Same

7 Are the estimated standard errors targeting the finite or super population

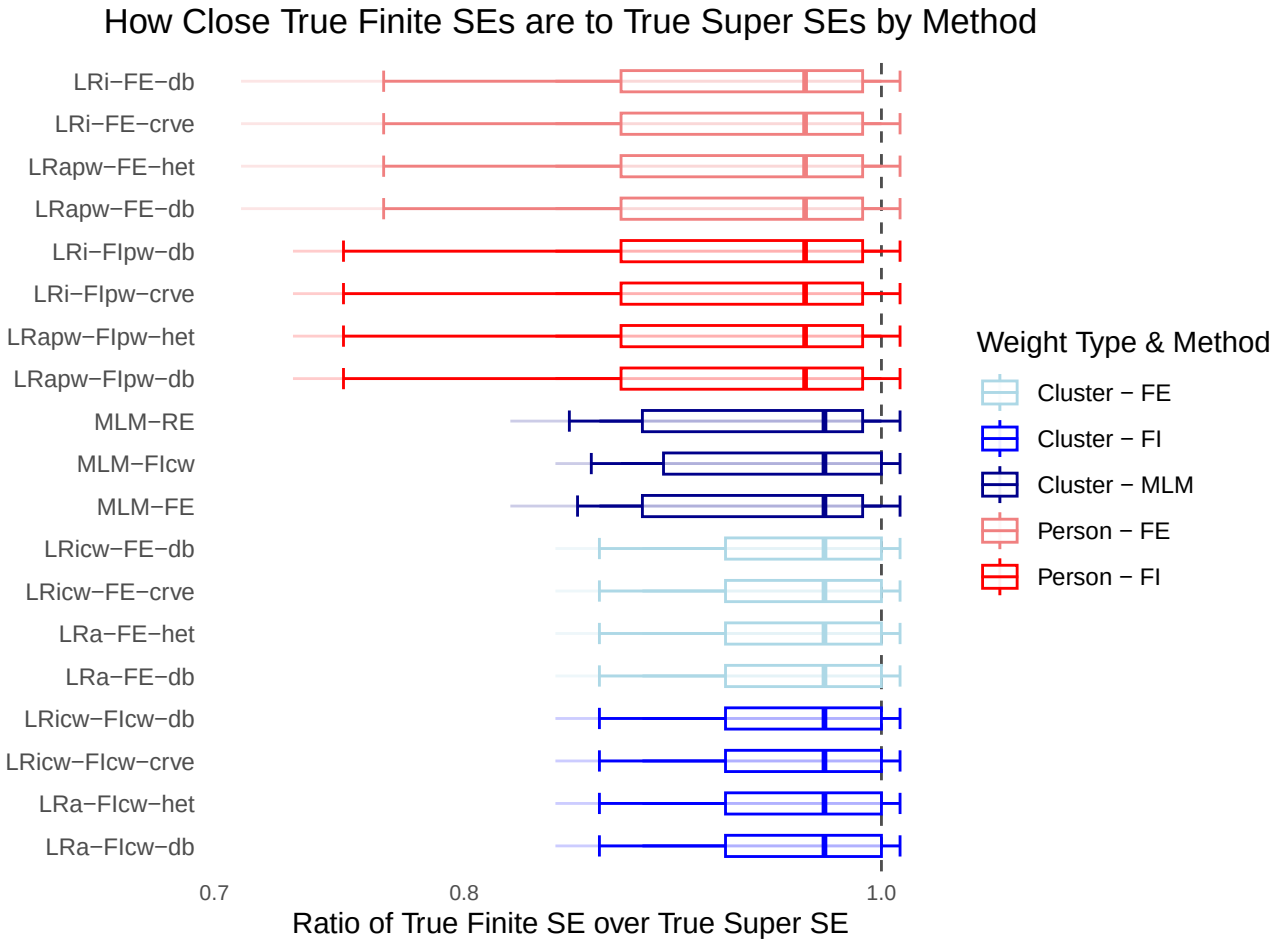
7.1 Main Bullet Points:

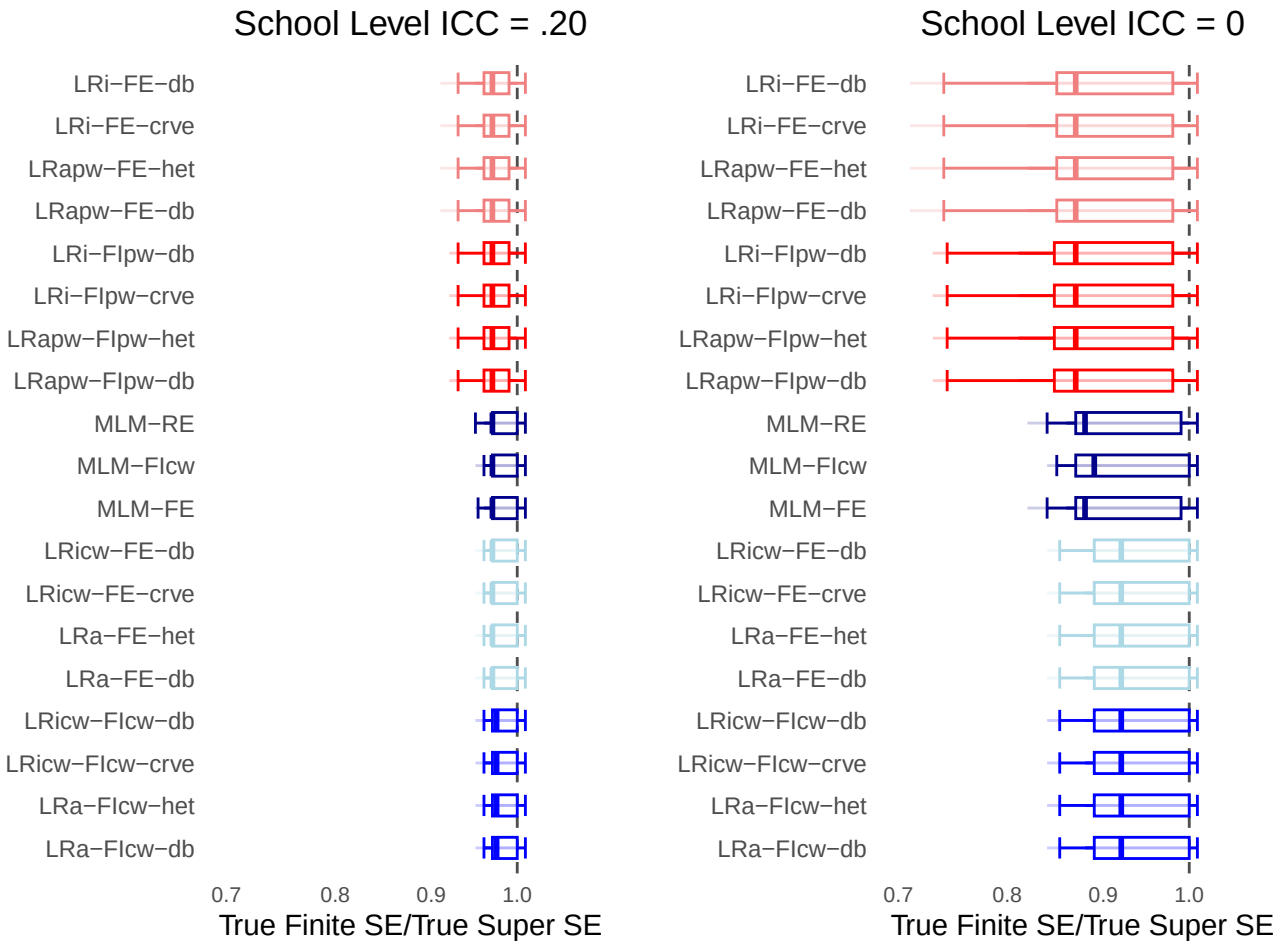
- This section answers the questions on page 8.
- As long as the school level ICC isn't zero, than the true finite and super population standard errors are very similar to each other (graph 2).

- I was having a little trouble visualizing whether the standard errors were targeting the finite population or super population standard errors, so I made up this metric: difference in % inaccuracy. This is calculated as the magnitude of ((Mean Estimated SE - True Super Population SE) / True Super Population SE) x 100 minus the magnitude of ((Mean Estimated SE - True Finite Population SE) / True Finite Population SE) x 100. When this number is positive the estimated standard errors are closer to the true super population standard errors and when this number is negative the estimated standard errors are closer to the finite population.
- The only estimators with standard errors that are ever really significantly closer to the finite standard errors than the super population standard errors are the person weighted fully interacted ones and the fully interacted MLMs (graphs three and four). It does look like the person weighted fully interacted estimators are targeting the finite standard errors when the ICC is zero, the blocks are very dissimilar, and there is high variance in the cluster sizes. Otherwise these estimators are also targeting super population standard errors (graph five).

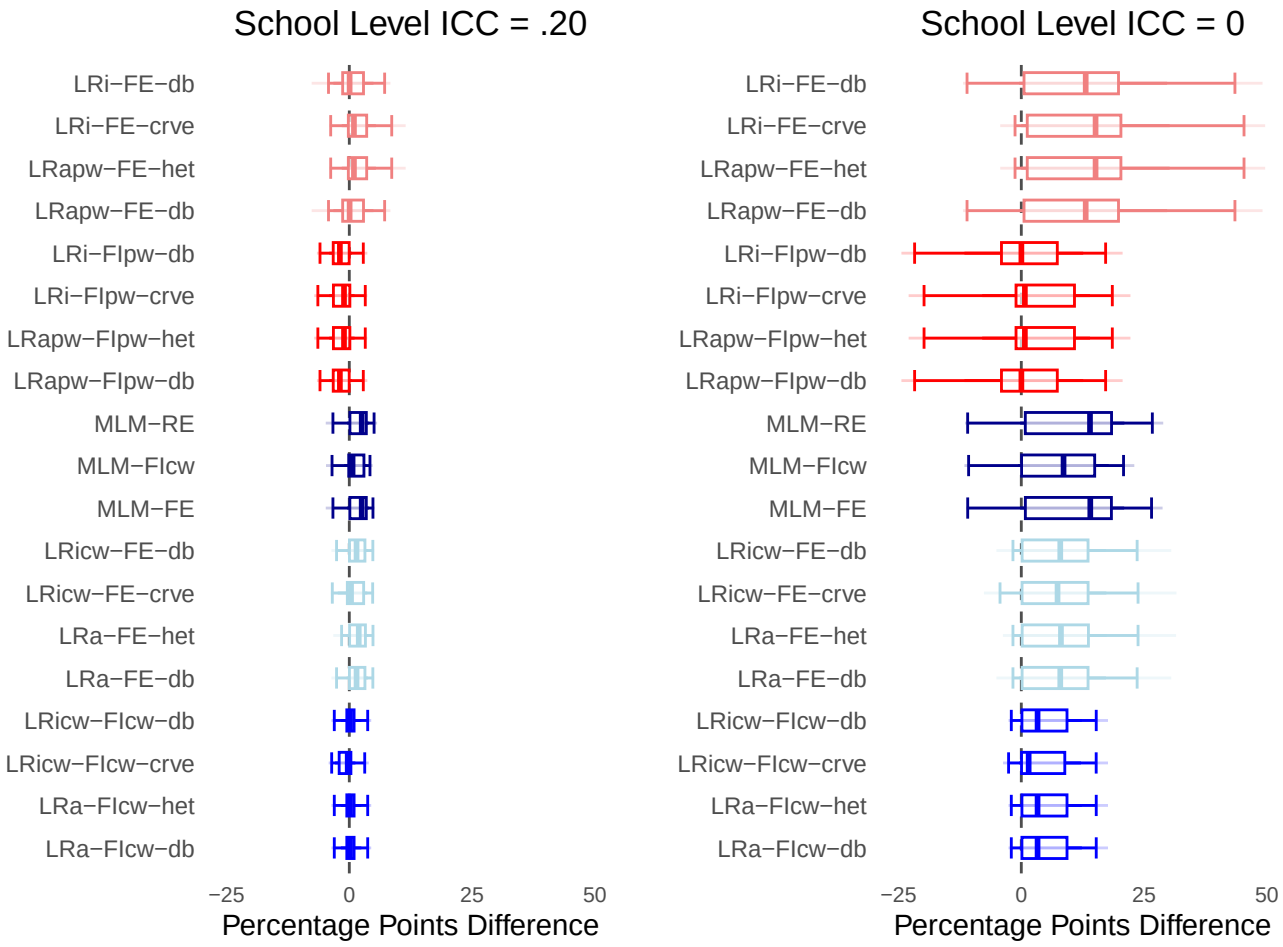
Description of Analysis:

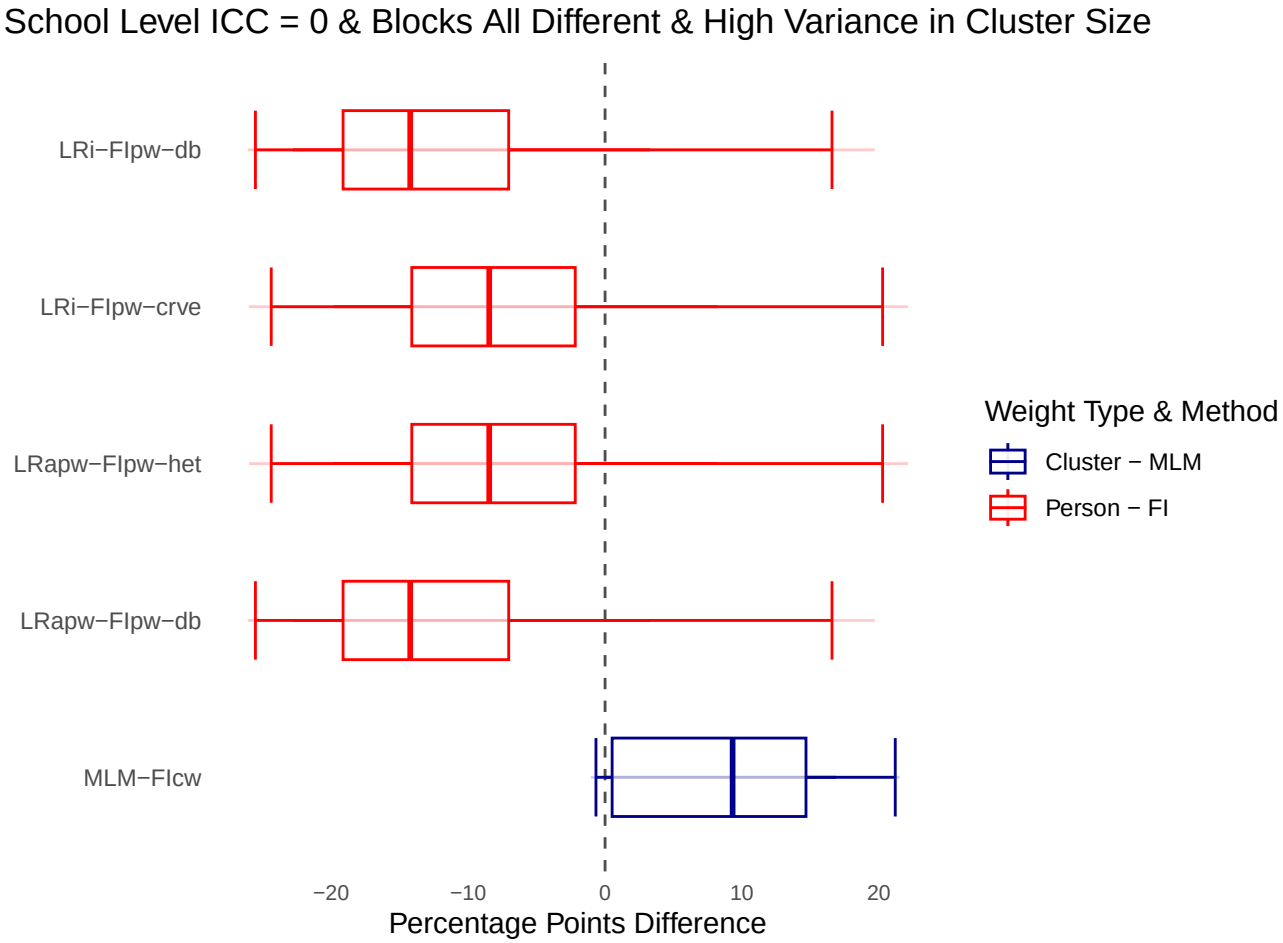
- The first graph is a boxplot of ratio of the true finite standard errors over the true super population standard errors across the different scenarios by method. When this metric is one the two standard errors are the same, when it's above one the finite population standard errors are bigger and when it's below one the super population standard errors are bigger.
- The second graph is the same as the first but broken out by ICC. In the graph on the left the school level ICC is .20 and on the right it's zero.
- The third graph is a boxplot of the difference in % inaccuracy across methods. This is calculated as the magnitude of ((Mean Estimated SE - True Super Population SE) / True Super Population SE) x 100 minus the magnitude of ((Mean Estimated SE - True Finite Population SE) / True Finite Population SE) x 100. When this number is positive the estimated standard errors are closer to the true super population standard errors and when this number is negative the estimated standard errors are closer to the finite population.
- The fourth graph is the same as the third but broken out by ICC. In the graph on the left the school level ICC is .20 and on the right it's zero.
- The fifth graph is a boxplot of the difference in % inaccuracy across methods, but only for the methods that might be targeting the finite population standard errors and only when the school level ICC is zero, the blocks are all messy, and the cluster size coefficient of variation is large.
- The table contains the five DGPs that produce the largest difference in % inaccuracy for each method, conditional on the magnitude being at least five percentage points.


How Close True Finite SEs are to True Super SEs by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
0.7
0.8
1.0
Ratio of True Finite SE over True Super SE
Weight Type & Method
Cluster − FE
Cluster − FI
Cluster − MLM
Person − FE
Person − FI


School Level ICC = .20
School Level ICC = 0
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
0.7
0.8
0.9
1.0
True Finite SE/True Super SE


How Much Closer the Estimated SE is to the True Super SE by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
−20
0
20
40
60
Percentage Points Difference
Weight Type & Method
Cluster − FE
Cluster − FI
Cluster − MLM
Person − FE
Person − FI


School Level ICC = .20
School Level ICC = 0
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
−25
0
25
50
Percentage Points Difference



7.1.1 Top Five Scenarios with Where Estimated SEs are Closest to Targeting Finite SEs

method	dif	infla te.f	infla te.s	true se.ratio	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	-5.07	1.01	0.936	0.92	S	+	0	L	0	+	Same
LRapw-FE-db	-11.9	0.975	0.856	0.88	S	+	0	L	S	0	Same
LRapw-FE-db	-11.7	0.976	0.859	0.88	S	+	0	L	L	+	Same
LRapw-FE-db	-11.7	0.975	0.858	0.88	S	+	0	L	L	0	Same
LRapw-FE-db	-11.5	0.973	0.858	0.88	S	+	0	L	0	0	Same
LRapw-FE-db	-11.4	0.965	0.851	0.88	S	+	0	L	S	+	Same
LRicw-FE-crve	-7.6	0.997	0.921	0.92	S	+	0	L	0	+	Same
LRicw-FE-crve	-7.43	1	0.926	0.93	S	+	0	L	L	+	Same
LRicw-FE-crve	-6.98	1	0.929	0.93	S	+	0	L	L	0	Same
LRicw-FE-crve	-6.73	0.996	0.929	0.93	S	+	0	L	0	0	Same
MLM-FE	-11.3	1	0.883	0.88	S	+	0	L	S	0	Same
MLM-FE	-11.2	1	0.887	0.89	S	+	0	L	S	+	Same
MLM-FE	-11.2	0.999	0.887	0.89	S	+	0	L	L	0	Same

method	dif	infla te.f	infla te.s	true se.ratio	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
MLM-FE	-11.2	0.996	0.884	0.89	S	+	0	S	L	0	Same
MLM-FE	-11	0.998	0.889	0.89	S	+	0	0	S	0	Same
MLM-FIcw	-11.6	0.99	0.874	0.88	S	+	0	S	S	+	Same
MLM-FIcw	-11.5	0.984	0.869	0.88	S	+	0	0	L	0	Same
MLM-FIcw	-11.4	0.982	0.868	0.88	S	+	0	L	S	0	Same
MLM-FIcw	-10.8	0.985	0.877	0.89	S	+	0	0	0	0	Same
MLM-FIcw	-10.7	0.97	0.863	0.89	S	+	0	S	L	0	Same
MLM-RE	-11.5	1	0.883	0.88	S	+	0	L	S	0	Same
MLM-RE	-11.2	0.997	0.885	0.89	S	+	0	L	S	+	Same
MLM-RE	-11.2	0.998	0.886	0.89	S	+	0	L	L	0	Same
MLM-RE	-11.2	0.995	0.884	0.89	S	+	0	S	L	0	Same
MLM-RE	-11	0.998	0.889	0.89	S	+	0	0	S	0	Same
LRapw-FIpw-db	-26.1	0.981	0.72	0.73	S	+	0	L	L	0	Mess
LRapw-FIpw-db	-25	0.999	0.748	0.75	M	+	0	L	S	0	Mess
LRapw-FIpw-db	-25	0.944	0.694	0.74	L	0	0	L	S	0	Mess
LRapw-FIpw-db	-24.4	0.914	0.67	0.73	S	+	0	L	0	+	Mess
LRapw-FIpw-db	-23.2	0.937	0.706	0.75	S	+	0	L	0	+	Mess
LRapw-FIpw-het	-26	0.982	0.722	0.74	L	0	0	L	S	0	Mess
LRapw-FIpw-het	-22.9	1.02	0.749	0.73	S	+	0	L	0	+	Mess
LRapw-FIpw-het	-21.3	1	0.786	0.79	S	+	0	L	L	0	Mess
LRapw-FIpw-het	-21.3	1.02	0.767	0.75	S	+	0	L	0	0	Mess
LRapw-FIpw-het	-20.8	1.01	0.787	0.78	S	+	0	L	L	+	Mess

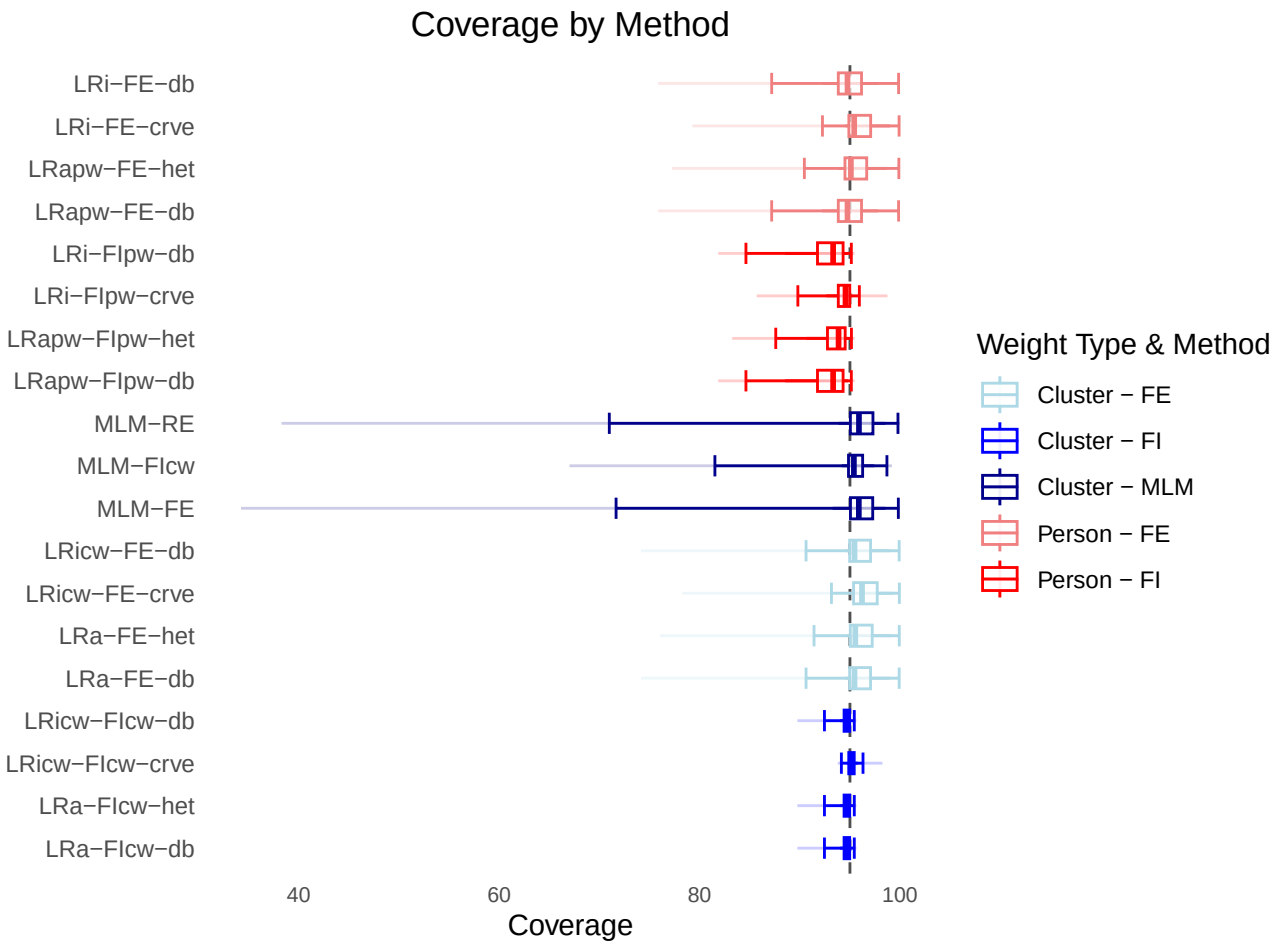
8 What do the confidence intervals look like for each estimator

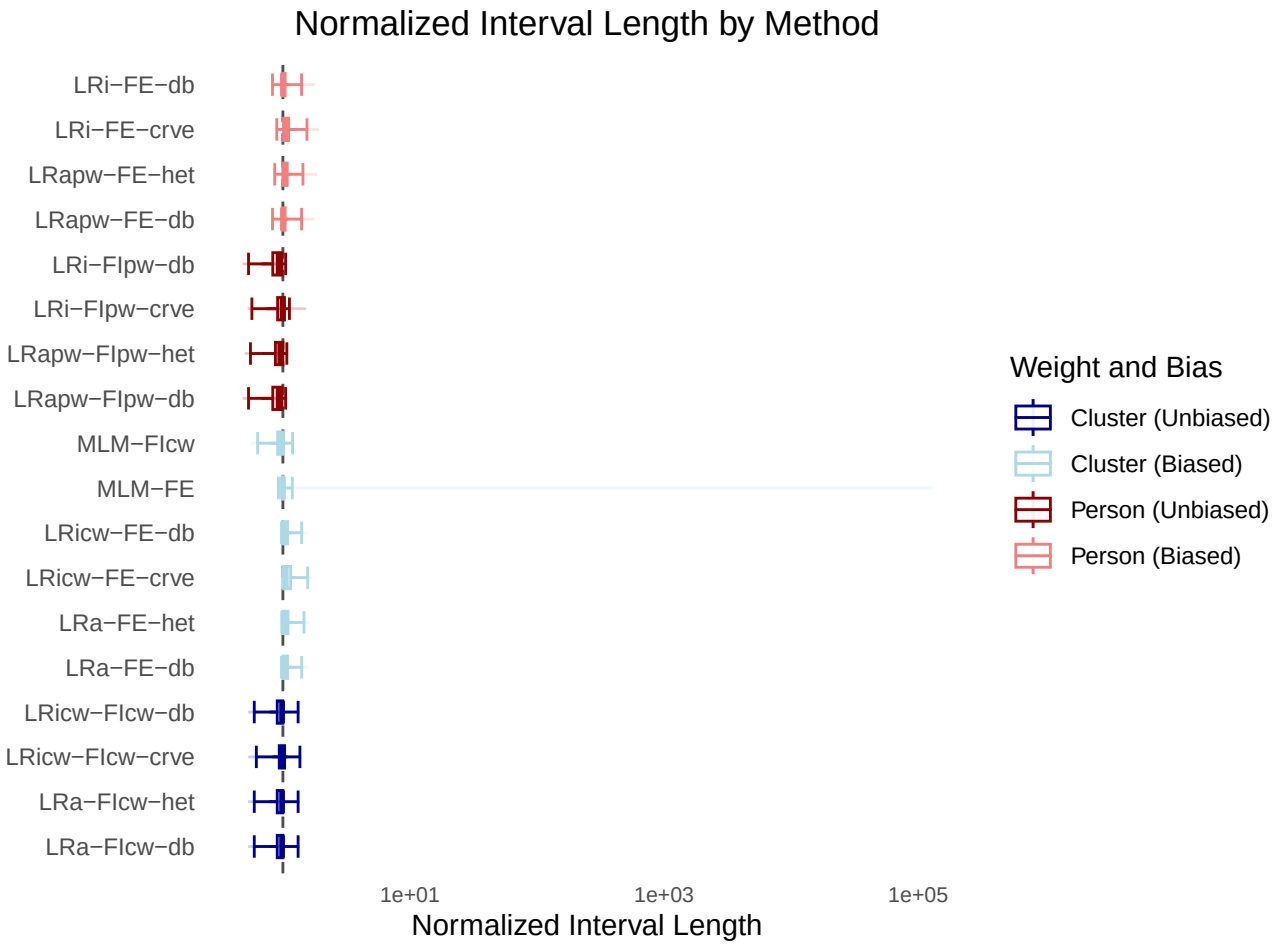
8.1 Main Bullet Points

- Coverage is the percentage of simulation replications where the estimator is within the 95% confidence interval. We would like the number to be 95% (the boxplots have a line there). The notable things are that the coverage for the MLM estimators are often very bad but other than that the coverage for the cluster weighted estimators are generally good. The coverage rates for the person weighted estimators however are pretty bad across methods, it's either very low (meaning the SEs are being underestimated) or too high (meaning the SEs are being overestimated within the same methods.

Description of Analysis:

- The first graph is a boxplot of the coverage by method. Coverage is the percentage of treatment effect estimates that fall in the 95% confidence interval. The vertical line is at 95%, which is what coverage should be.
- The second graph is a boxplot of the normalized 95% confidence interval length by method.


Coverage by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
40
60
80
100
Coverage
Weight Type & Method
Cluster − FE
Cluster − FI
Cluster − MLM
Person − FE
Person − FI



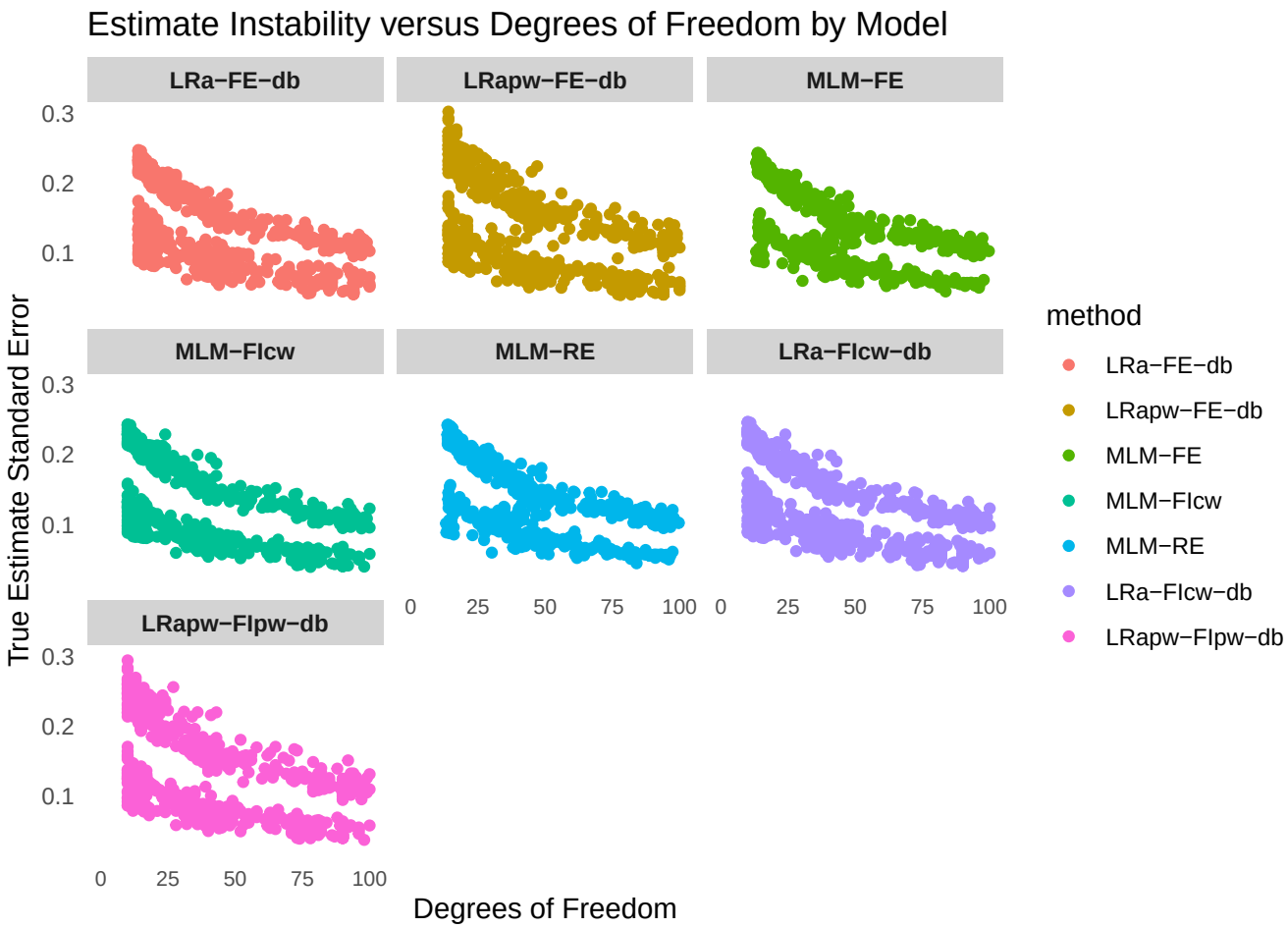
9 Is Low Degrees of Freedom Causing Problems because of Estimate Instability Driven or Bias

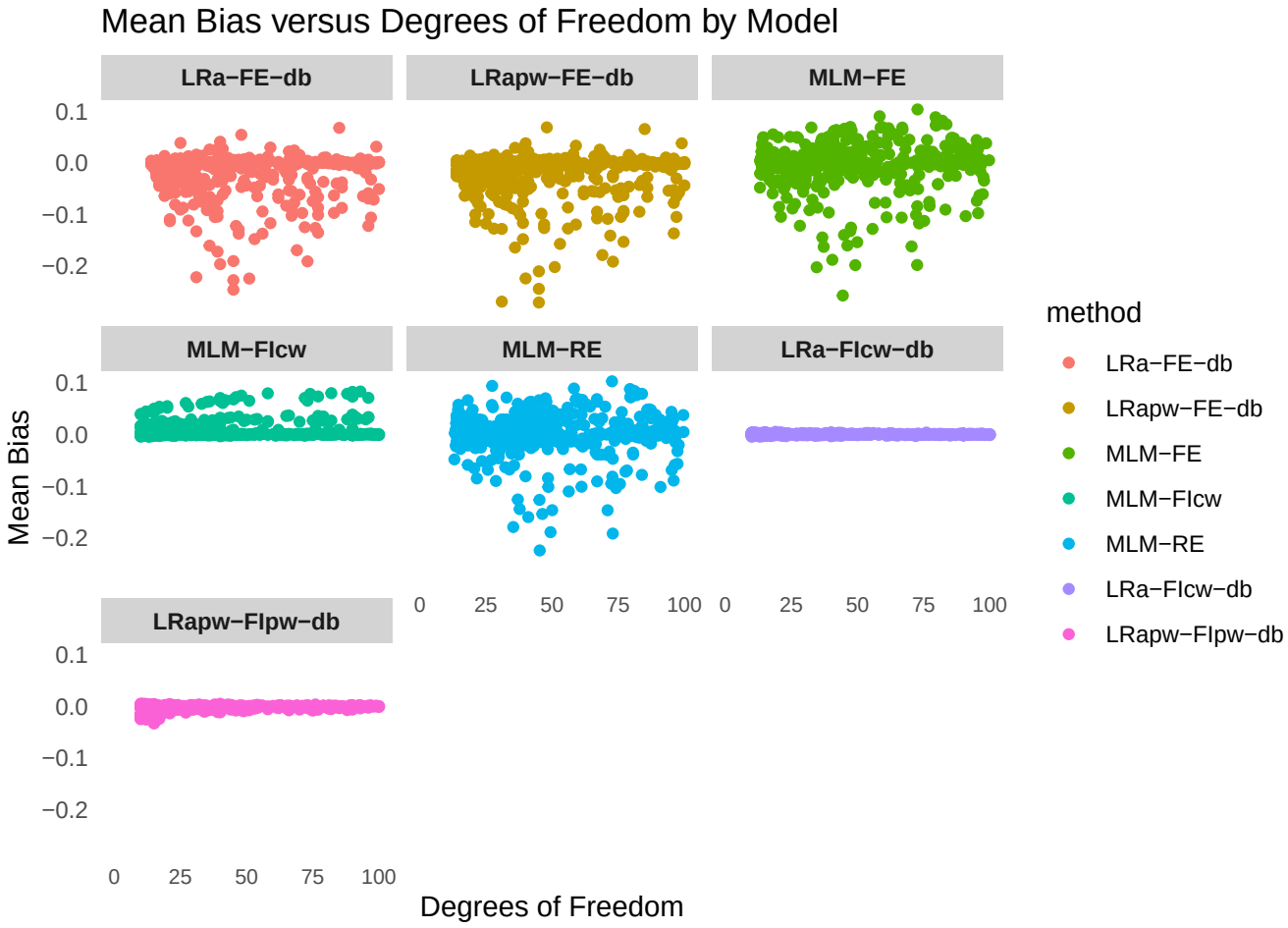
9.1 Main Bullet Points

- This is answering the question on page 35
- Low degrees of freedom is clearly causing problems because of estimate instability. There is clear increase in estimate instability for the low degrees of freedom DGPs (graph one) and no relation between degrees of freedom and mean bias (graph two).

9.2 Description of Analysis:

- The first graph is scatter plot of the true estimate standard error across replications against the model degrees of freedom, graphed sepparately for each method
- The second graph is scatter plot of the mean bias across replications against the model degrees of freedom, graphed sepparately for each method


Estimate Instability versus Degrees of Freedom by Model
LRa−FE−db
LRapw−FE−db
MLM−FE
MLM−FIcw
MLM−RE
LRa−FIcw−db
LRapw−FIpw−db
True Estimate Standard Error
0.3
0.2
0.1
0
25
50
75
100
Degrees of Freedom
method
LRa−FE−db
LRapw−FE−db
MLM−FE
MLM−FIcw
MLM−RE
LRa−FIcw−db
LRapw−FIpw−db


Mean Bias versus Degrees of Freedom by Model
LRa−FE−db
LRapw−FE−db
MLM−FE
MLM−FIcw
MLM−RE
LRa−FIcw−db
LRapw−FIpw−db
0.1
0.0
−0.1
−0.2
0
25
50
75
100
Mean Bias
Degrees of Freedom
method
LRa−FE−db
LRapw−FE−db
MLM−FE
MLM−FIcw
MLM−RE
LRa−FIcw−db
LRapw−FIpw−db